

Towards the Statistical Inference of Realistic Motor Commands: Designing Neuromechanical Models to Bridge Large-Scale Neuronal and Behavioral Data on Zebrafish Locomotion

Honors Thesis

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Contents

1	Introduction	2
2	Background	4
2.1	Zebrafish Neurobiology	4
2.2	Latent Dynamical Models	5
3	Results	7
3.1	Zebrafish Tail Motion Exhibits Expected Spatial and Temporal Properties	7
3.2	Neural Network Models	8
3.2.1	Model Setup	8
3.2.2	Specific neural network Parameters are Required to Produce Oscillatory Movement . .	11
3.2.3	Biomechanics are necessary to produce realistic swim bouts	11
3.3	Biomechanical models	12
3.3.1	Model Setup	12
3.3.2	Eigenanalysis of Biomechanical Models Gives Conditions to Produce Oscillations . . .	13
3.3.3	Resulting Temporal Frequencies May Lead to Inaccurate Motion	14
3.3.4	Broadly Distributed Motor Inputs are Likely Necessary	15
3.3.5	Biomechanical Model Parameters Have Large Variable Effects on Tail Motion	16
3.4	Inference of biomechanics from data	19
3.4.1	Linear Regression Gives Intuition Behind Biomechanics Parameters	19
3.4.2	Regression Parameters and Torques Produce Implausible Tail Locomotion	20
3.5	Towards inference of neuro-biomechanical systems	22
3.5.1	iLQR Demonstrates Observation-Dependent Dynamics May Not Be Unique	22
4	Discussion	24
4.1	Capabilities and Limitations of Our Biologically Relevant Model	24
4.2	Likelihood of ReLU RNNs as a model for locomotor circuits	25
4.3	Neuron control of muscle stiffness and neuromusculoskeletal models	26
4.4	Towards a full brain-body-environment model	27
4.5	Parameter Tuning	28
5	Conclusion	30
6	Acknowledgments	31
	References	32
7	Methods	37
7.1	Zebrafish Swim Bout Data	37
7.2	Spatial Wavelength and Temporal Phase Lag	37
7.3	Linear Regression	37
7.4	iLQR: Iterative Linear Quadratic Regulator	38

1 Introduction

Locomotion is a ubiquitous, evolutionarily conserved behavior throughout the animal kingdom.^{1–3} The neural circuits controlling hindlimbs is highly dependent on the physics and structure of the animal,⁴ and likely co-evolved with body dynamics and constraints to account for these interactions.⁵ Motor commands therefore cannot be accurately inferred without a realistic neuromechanical model design. Numerous biomechanical frameworks for studying locomotion have been proposed that incorporate both neural and body mechanics,^{6–8} all with various levels of complexity. These models in computational neuroscience have given insights into how locomotion is controlled between the brain, body, and environment in a unified control of behavior.^{9–17} Despite this progress, the neural control of locomotor behaviors is not yet fully understood.¹⁸ It remains unclear how motor control inputs are generated and how neural circuits process these inputs to create locomotive movements.

To study these underlying circuits, neuroscientists have turned to the zebrafish for its intriguing capability in large-scale data analysis. The zebrafish have become an emerging model organism in biology and neuroscience.¹⁹ Their behavioral and neurological properties have made them an appealing model in understanding locomotor control in vertebrates. Larvae zebrafish consist of mostly a head and a tail, and move in the water through quick bouts of their tail. In recent years, this locomotor repertoire has been extensively studied.^{20–22} Specific²¹ and sparse^{18,23,24} motor control inputs in the reticulospinal system controls these swim bouts, and each bout exhibits specific kinematics in order to perform a movement (left turn, right turn, etc.). This behavioral data provides an extensive tool for data analysis of locomotion. In addition, the zebrafish possess exciting opportunities in neurological data. The transparent body of larvae zebrafish coupled with calcium-imaging technology makes their neural activity optically accessible,¹⁹ allowing for whole-brain recordings during behaviors.

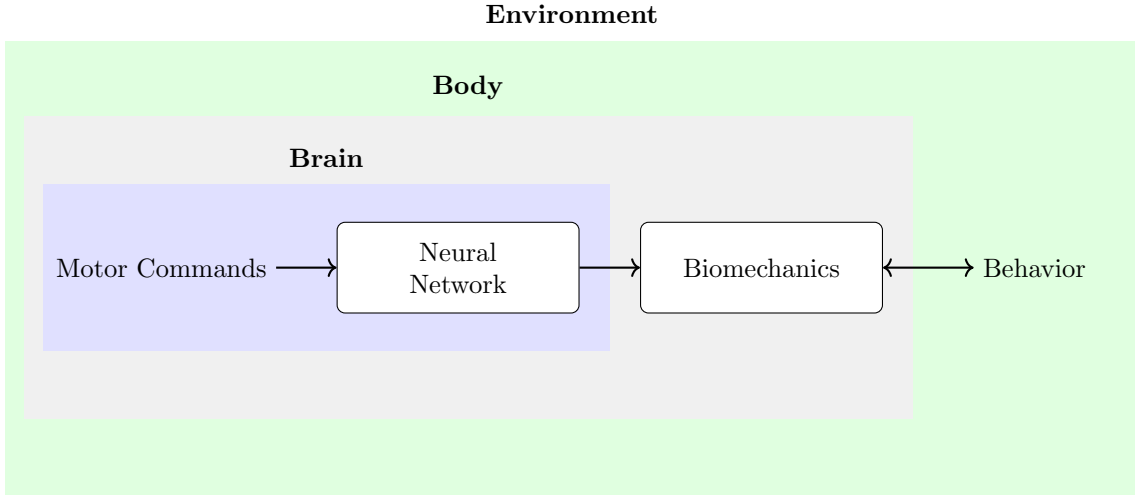


Figure 1.1: Proposed neuromechanical framework modeling interactions between the brain, body, and environment. Motor commands are sent to a model for neural activity, which outputs into a biomechanical model of physical forces. Interactions between the body and environment then influences resulting behavior.

A sufficient biomechanical architecture is necessary to bridge the gap between these large behavioral and neurological datasets. One appealing and fast-growing option is the neural network, a revolution in data science computing capabilities. Arising as a computational model of neuron interactions in the brain, neural networks have been a popular model in computational neuroscience to understand how and why neural circuits interact the way they do. Recurrent neural network models have been shown to explain a variety of complex behaviors, including vision,²⁵ memory,²⁶ and decision making.^{27–29} However, neural networks in the realm of data science have only been explored in terms of increasing model performance. Current standard artificial neural networks such as minimal gated unit recurrent neural networks have better performance than traditional neural networks but have lost much of their biological relevance and plausibility. This makes artificial neural networks problematic to use as a viable architecture to study realistic behavior.

Indeed, while they have been shown to be accurate in describing behavioral data,¹⁵ its internal dynamics are less meaningful to the actual neuronal dynamics of the behavior it describes.

This project aims to utilize cutting edge methods of machine learning to describe neuronal behavior in zebrafish while using biologically plausible networks. The resulting trained dynamics will then be relevant to describe actual dynamics in the zebrafish nervous system. A key question we seek to understand, which prevents us from a more traditional black box training approach, is what the brain is optimizing in order to encode these swim bouts into motor control inputs. In our research, we built towards a biomechanical model that incorporates both neural activity in the zebrafish spinal cord and the affect of tail motion on the zebrafish body in the water according to Newton’s laws of motion. Then, we used various inference methods to understand how swim bouts are controlled via model inputs and parameters. These inferences gave insights into how locomotor behavior is likely controlled. The result is information on optimal dynamics the zebrafish brain uses to perform locomotor behaviors.

In section 2, we explain some of the neurobiological background of the zebrafish locomotor repertoire and behavioral organization, as well as the dynamical framework used throughout our model development. In section 3.1, we replicate previous findings on vertebrate swimming behavior to establish a statistical estimate for comparing realistic swim bouts against modeled motion. In section 3.2, we set up our neural network model of the zebrafish spinal cord. We give specific conditions on the networks parameters that are necessary to replicate swimming oscillations. We also demonstrate that these oscillatory movements are not enough to replicate realistic swimming alone. In section 3.3, we extend our neurological model to a neuromechanical framework by assuming biomechanical equations governing zebrafish locomotion. We exploit the linearity in our model to establish additional parameter constraints, and use these constraints to gain further insights into biomechanical behavior. Next, we explore the underlying parameter space using simulations of artificial locomotion. These insights prime us to then fit behavioral data to our biomechanical equations via linear regression in section 3.4. This inference illuminated properties of the biomechanical parameters that are likely important in generating complex movement. Finally, in section 3.5 we worked towards understanding a full neuromechanical model using control-based inference through iLQR, which showcased that modeling locomotion based only on differences in neuron activity is likely not enough to generate unique locomotor inputs.

2 Background

2.1 Zebrafish Neurobiology

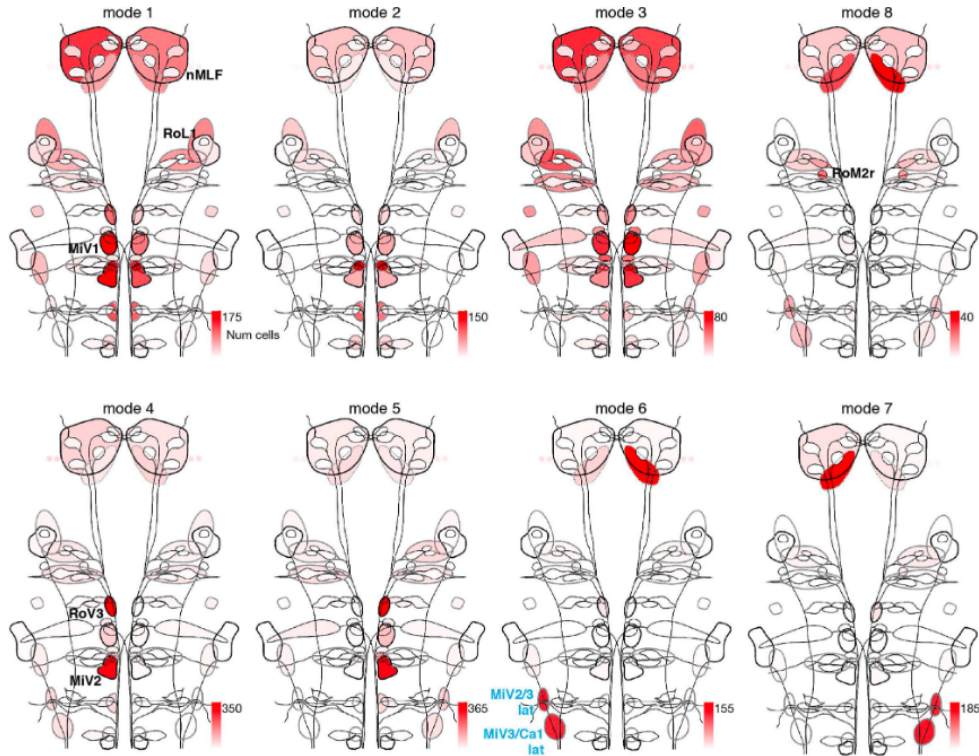


Figure 2.1: Identified eight motor command inputs sent from the zebrafish reticulospinal system, taken from²¹

Understanding how neural circuits generate movement is a fundamental question in neuroscience. Zebrafish, with their diverse locomotor behaviors, provide an ideal model for studying descending motor control. However, identifying the control inputs that drive these behaviors and how the nervous system interprets them remains challenging.

The part of the nervous system that controls locomotor movement in zebrafish is the reticulospinal system, which includes clusters of neurons in the brain, brain stem, and along the spinal cord in the tail.¹⁸ For locomotion, excitatory inputs are sent from the brain down the tail, and this generates tail movement.^{1,18} Previous research has identified various modes of neural activity in the reticulospinal system.^{21,22} These modes have activity distributed throughout the entire system and are not concentrated to single neurons (Figure 2.1). This low-dimensionality of organized *functional archetypes* is robust enough to determine tail behavior characteristics and is a common property seen in motor behavior.^{30,31} Additionally, it is behavior specific,²⁰ with previous research identifying five archetypes used to generate swim speeds and direction independently, and three used to generate specialized swims used for hunting.²¹

Zebrafish tail behavior is typically recorded in terms of the *cumulative tail angle* (Figure 2.2). If we imagine the tail as n distinct segments, each bending a certain amount θ (the *relative angle*), the cumulative angle is the sum of the relative angles up to that segment. This gives us a description of how different parts of the tail are bending over time during a swim bout. Another important measurement regarding tail behavior is spatial wavelength and temporal phase lag. Spatial wavelength is the measurement of how many wavelengths the fish beats their tail at, while temporal phase lag is the delay in movement between two different points on the tail in proportion to overall tail-beat frequency. These measurements have previously been shown to be roughly constant in lamprey throughout locomotor behavior.^{32–36} Control of these attributes is theorized to be an important property of realistic swimming motion.

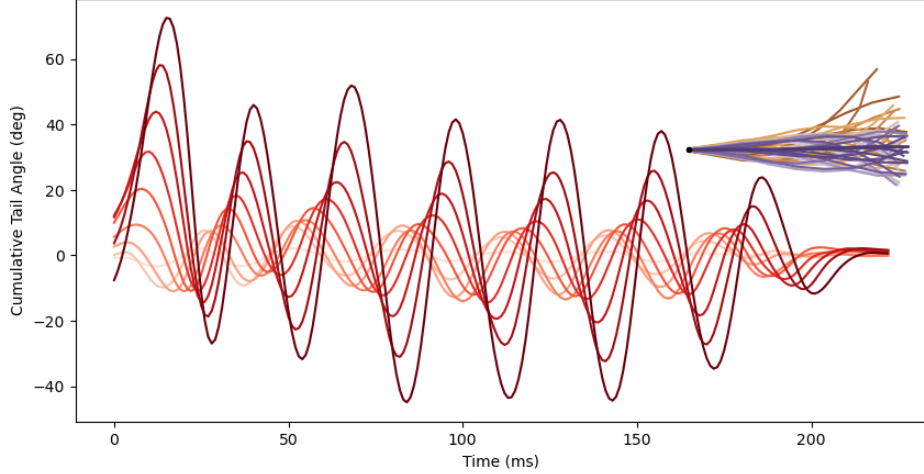


Figure 2.2: Example rightward forward swim bout. Darker red color indicates angle measurements from more caudal segments. Top right inset showcases motion of tail over time, with time varying from orange to purple.

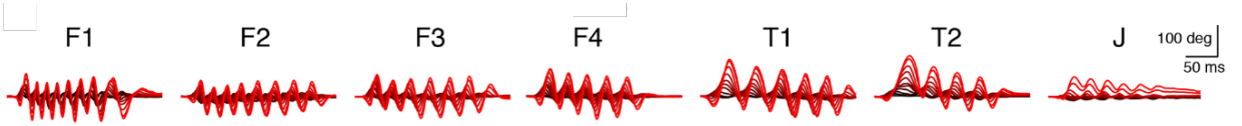


Figure 2.3: Example (right oriented) swim bout types of zebrafish, represented as cumulative tail angle plots.

Zebrafish movements have also been categorized similar to neurological data. Locomotor behavior can be divided into a kinematic sequence of seven (each with left and right oriented versions) swim bout types, sorted by amplitude of the cumulative angle over time.²¹ Four of these bouts represent forward swims, two for turning swims, and one for the characteristic J-turns in which the tail creates a J shape to orient the fish (Figure 2.3).

Why and how the reticulospinal system is organized in this manner to send motor input commands is unknown. While tail movement can be adequately explained by each of the functional archetypes, and these archetypes are clearly specialized for specific behaviors, it's unclear why motor commands are setup this way and how the spinal cord neural circuits integrate these inputs into motor movement. A data science perspective may offer a way to bridge this gap. By linking behavioral and neurological data through a complete biomechanical model of the zebrafish, we can understand how neurological inputs may lead to different locomotion. Data analysis gives us a toolkit to study how these organizations influence each other through inference methods on these models.

2.2 Latent Dynamical Models

To develop a model of zebrafish locomotor movement, we consider the latent dynamics of the reticulospinal system. Latent dynamics are the high-level hidden interactions underlying a complex high dimensional system. Modeling latent dynamics rather than each individual neuron can help describe behavior while allowing us to simplify the circuit's complexity. This framework follows previous methods in understanding the dynamics of neural circuits and resulting observations.^{15,21,37} We consider the following latent dynamical equations (Figure 2.4):

$$\begin{aligned} \text{latent dynamics } z_t &= f_\theta(z_{t-1}, u_t), \\ \text{outputs } \tau_{a_t} &\sim p_\theta(\tau_{a_t} | z_t), \\ \text{observations } \phi_t &= g(\phi_{t-1}, \tau_{a_t}). \end{aligned}$$

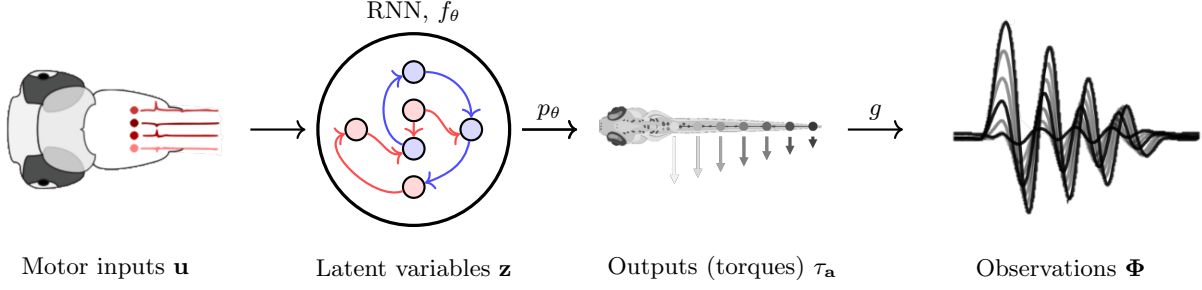


Figure 2.4: Illustration of model latent dynamical model setup for zebrafish locomotor movement.

Here, f is the biologically plausible spinal cord RNN model. It uses motor control inputs u_t and gives the neuronal latent dynamics z_t along the tail, for some instance of time t . The outputs produced by the neuronal dynamics (according to some probability distribution p_θ) are the active torques τ_a on the tail that generate the swim bout. Importantly, we do not actually observe these torques but rather observe the tail angles ϕ_t . We can get from the torques to the resulting tail angles by considering the biomechanics of the tail, which is our model g . This layout provides a structure for how we will develop our neural network and biomechanical models, and how they will interact with each other in a unified neuromechanical model.

3 Results

3.1 Zebrafish Tail Motion Exhibits Expected Spatial and Temporal Properties

Our data consists of various swim bouts of a specific zebrafish (Methods 7.1). Before we set out to model this data, we used it to study properties of realistic zebrafish movement to understand what our models needed to accomplish. We replicated results from previous studies on lamprey, which found characteristic constants during swimming.^{32–36} These properties were the wavelength of the tail oscillation, which we refer to as the *spatial wavelength*, and the timing difference between evenly spaced segments along the tail, which we refer to as the *temporal phase lag*. These properties remain constant for all locomotor speeds, and will allow us to establish a statistic metric of comparison for artificial swimming.

Spatial wavelength and temporal phase lag was measured for swim bouts in our dataset (Methods 7.2). For each swim bout, we obtained a mean wavelength estimate λ_μ and its variance overtime λ_σ , as well as a mean phase lag estimate ρ_μ and the intersegmental variance ρ_σ . To agree with previous results, we would expect similar λ_μ and ρ_μ throughout all swim bouts as well as $\lambda_\sigma, \rho_\sigma \ll 1$.

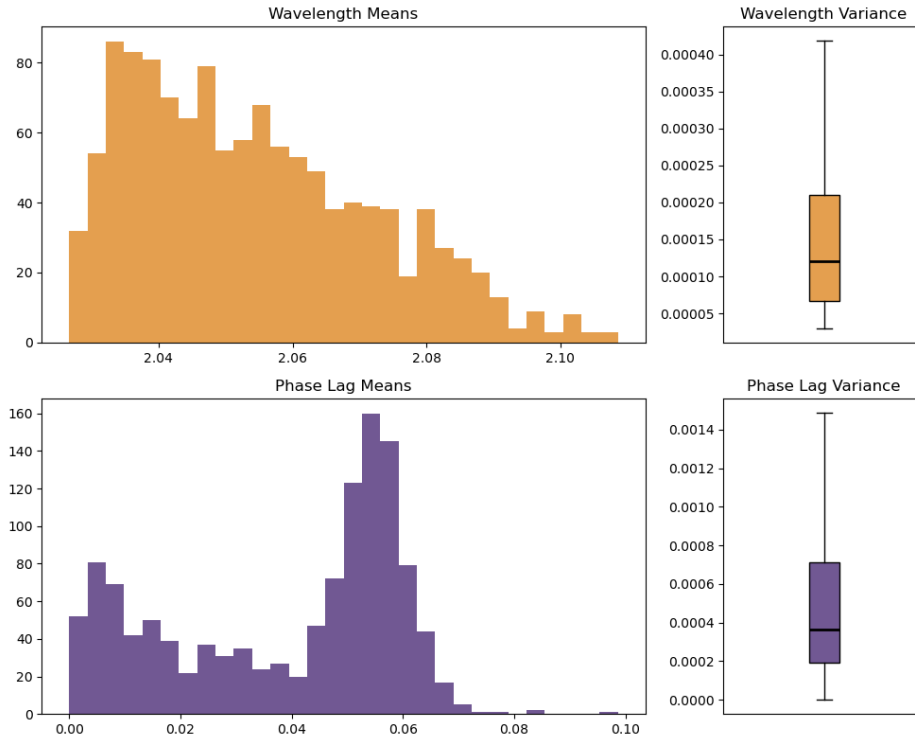


Figure 3.1: Top and bottom left plots, histograms of spatial wavelength means and phase lag means, respectively, over entire data set with outliers removed. Top and bottom right plots, box plots of wavelength variance and phase lag variance, respectively, over entire data set.

First, we analyzed the spread in each of these measurements over the entire dataset (Figure 3.1). Both the wavelength and phase lag had variances that were very close to 0 with small spreads, indicating that our zebrafish swim bouts had spatial wavelengths relatively constant in time, as well as relatively constant phase lag between segments. The distribution of spatial wavelength means is unimodal, while the distribution of phase lag means is more bimodal, indicating two possible “settings” in the control of phase lag depending on desired movement. Both have most of the distribution spread in a 0.05-0.1 unit range. This spread of wavelength and phase lag may also be attributed to swim bout type or swim bout speed. Nevertheless, the small distribution spreads indicate that these values stay constant within some error throughout locomotor motion.

Next, we looked at the wavelength and phase lag measurements grouped by swim bout type to control for

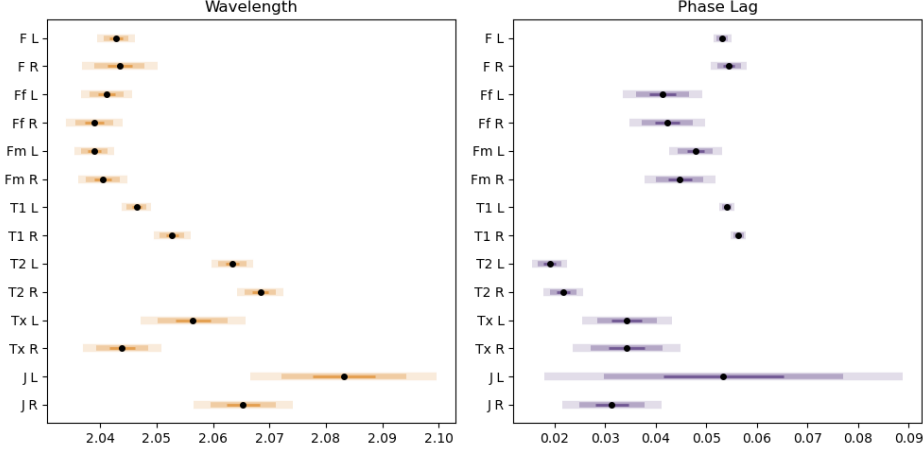


Figure 3.2: On left, distribution plot of λ_μ averaged across each swim bout type. On right, distribution plot of ρ_μ averaged across each swim bout type. Colored bars display CI intervals, with increasing darkness in color indicating 99.7%, 95% and 68% CI intervals.

differences in locomotor behavior. We found that each swim bout type produce roughly the same wavelength (a little greater than 2) with variations between, but small CI intervals within each swim bout type. CI intervals for forward swims overlapped, indicating confidence in a shared estimate of spatial wavelength for these swims. However, turns and J-turns exhibited greater variance and mostly did not overlap with forward swims. We believe this is due to their complex and asymmetric behavior. Hence, the spread in distribution from Figure 3.1 is likely more attributed to differences in swim bout type, rather than variance within a swim bout type. These results also show that there is large variance in phase lag both between and within swim bout types. J turns seem to have the largest variance within their group, while some turns exhibit the lowest phase lag overall. This may be attributed to different types of locomotor coordination needed when initiating escape swims or turns. Indeed, previous research shows that phase lag is no longer constant throughout the body for non symmetrical forward swims in lamprey.³⁴ This likely also explains why more complicated forward swims or turning swims exhibit greater variance in phase lag. Focusing on forward swims, there still seems to be some variance between swim types, but less variance within each type. Therefore, we reason that phase lag can still be a meaningful property of forward swims, within some range of 0.05-0.07.

Our results show that wavelength and phase lag are roughly constant throughout locomotor behavior, lining up with previous results. However, it is evident that variation still exists within these estimates, likely attributed to differences in locomotor control to execute more complicated swims. Hence, these properties give a meaningful description of the aspects of different swim bouts. Simultaneously, the relatively low variation in some of these estimates provides us with a method of verification for whether artificial swimming behavior lines up with realistic motion. Therefore, we will look at measurements of wavelength and phase lag to get a sense of how realistic our models may be. To mitigate as much error as possible in these estimates, we focused on basic rightward and leftward forward swims (F_R and F_L swims). We will also rely on visual information to observe our findings to account for the errors in wavelength and phase lag estimates.

3.2 Neural Network Models

3.2.1 Model Setup

To work towards establishing our neuromechanical model, we began by focusing on the neural dynamics alone. We used previous models of neural circuits for swimming to establish a set of equations governing a biologically realistic neural network model of the zebrafish spinal cord. This model will be the foundation for understanding what is required to generate complex, oscillatory behavior in swimming. To design such a network with biological realism, we will use the neurobiology of the zebrafish reticulospinal system to

determine how our neural network is organized.

Before we incorporate neurobiology into our design, we familiarize the reader with the typical equations for a firing rate neural network in computational neuroscience. The firing rate of neuron populations in neural circuits can be modeled by the recurrent neural network design

$$\tau \cdot \dot{\mathbf{z}} = -\mathbf{z} + f(W\mathbf{z} + W_u\mathbf{u} + \mathbf{b})$$

where neural activity (ie. the firing rate of the neuron population) is given by $\mathbf{z}$, inputs given by $W_u\mathbf{u}$, $\mathbf{b}$ is a resting firing rate vector, τ is a time constant vector, f is a non-linear activation function, and W is known as the *synaptic weight matrix*. Connections between neurons are represented as values in this matrix. Note both $\mathbf{z}$ and $\mathbf{u}$ vary with time, and the leak term $-\mathbf{z}$ models the neural activity dynamically evolving between steady states. Hence this is a dynamic model of the neural activity over time subject to time-varying inputs and constant recurrent connections.

In zebrafish, reticulospinal neurons send motor inputs through excitatory synapses to spinal neurons. These inputs are integrated by spinal motoneurons that concentrate at segments along the tail to generate movement. These neurons follow an oscillatory pattern, and this is typically modeled through *central pattern generators* (CPGs) which are neural circuits that generate different types of oscillations.^{32,38–44} For now, we have chosen to model the spinal motor neuron network using linearly coupled E-I oscillators with descending commands (Figure 3.3), which is a typical network design for CPGs. Our network follows similar past models of a 2 interneuron design,^{13,43,44} in which each layer represents a segment along the tail and is made up of a pair of excitatory and inhibitory neurons on both the left and right of the spine. We then include ipsilateral excitatory projections (ie. connections of positive weight) from excitatory neurons and both ipsilateral and contralateral inhibitory projections (ie. connections of negative weight) from inhibitory neurons on each side of the spinal cord. Each new layer then receives input from excitatory neurons in the previous layer.

In terms of our model, this gives (using n layers)

$$\mathbf{z} = \begin{bmatrix} E_{L_1} \\ I_{L_1} \\ I_{R_1} \\ E_{R_1} \\ \vdots \\ E_{L_n} \\ I_{L_n} \\ I_{R_n} \\ E_{R_n} \end{bmatrix}, \quad W = \begin{bmatrix} 0 & -w'_{ips} & -w'_{con} & 0 & 0 & 0 & 0 & 0 & \dots \\ w_{ips} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & w_{ips} & 0 & 0 & 0 & 0 & \dots \\ 0 & -w'_{con} & -w'_{ips} & 0 & 0 & 0 & 0 & 0 & \dots \\ w_{desc} & 0 & 0 & 0 & 0 & -w'_{ips} & -w'_{con} & 0 & \dots \\ 0 & 0 & 0 & 0 & w_{ips} & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & w_{ips} & \dots \\ 0 & 0 & 0 & w_{desc} & 0 & -w'_{con} & -w'_{ips} & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

where $w_{ips}, w'_{ips}, w'_{con}, w_{desc} > 0$. Note we have assumed each layer has the same synaptic constants. Additionally, for simplicity, we have assumed that the synaptic weight of descending commands w_{desc} is constant for each segment. From this, we have that the first layer is governed by the following differential equations:

$$\begin{aligned} \tau_e \frac{dE_{L_1}}{dt} &= -E_{L_1} + f(-w'_{ips}I_{L_1} - w'_{con}I_{R_1} + w_{ue}u_L + b_e), \\ \tau_i \frac{dI_{L_1}}{dt} &= -I_{L_1} + f(w_{ips}E_{L_1} + w_{ui}u_L + b_i), \\ \tau_i \frac{dI_{R_1}}{dt} &= -I_{R_1} + f(w_{ips}E_{R_1} + w_{ui}u_R + b_i), \\ \tau_e \frac{dE_{R_1}}{dt} &= -E_{R_1} + f(-w'_{ips}I_{R_1} - w'_{con}I_{L_1} + w_{ue}u_R + b_e), \end{aligned}$$

while the remaining layers are governed by the equations

$$\begin{aligned}\tau_e \frac{dE_{L_k}}{dt} &= -E_{L_k} + f(-w'_{ips} I_{L_k} - w'_{con} I_{R_k} + w_{desc} E_{L_{k-1}} + b_e^{(desc)}), \\ \tau_i \frac{dI_{L_k}}{dt} &= -I_{L_k} + f(w_{ips} E_{L_k} + b_i), \\ \tau_i \frac{dI_{R_k}}{dt} &= -I_{R_k} + f(w_{ips} E_{R_k} + b_i), \\ \tau_e \frac{dE_{R_k}}{dt} &= -E_{R_k} + f(-w'_{ips} I_{R_k} - w'_{con} I_{L_k} + w_{desc} E_{R_{k-1}} + b_e^{(desc)}),\end{aligned}$$

where we have chosen f to be the ReLU activation function given by $f(x) = \max\{0, x\}$. Hence, this is a ReLU RNN model. The values $b_e, b_e^{(desc)}$, and b_i are chosen so that each neuron has a given resting state. Given resting states $E_L, E_R = r_e$ and $I_L, I_R = r_i$ we can solve for these constants in the first layer by assuming zero inputs and setting each derivative to 0:

$$\begin{aligned}b_e &= r_e + r_i(w'_{ips} + w'_{con}), \\ b_i &= r_i - r_e(w_{ips}).\end{aligned}$$

Similarly, we find the excitatory constant for each additional alyer:

$$b_e^{(desc)} = b_e - r_e w_{desc}$$

Following our latent dynamical framework, we then model the active torque on the tail according to a normal distribution determined by the difference in excitatory neuron activity:

$$\tau_{a_k} \sim N(a(E_{R_k} - E_{L_k}), \sigma^2),$$

where $a, \sigma^2 > 0$.

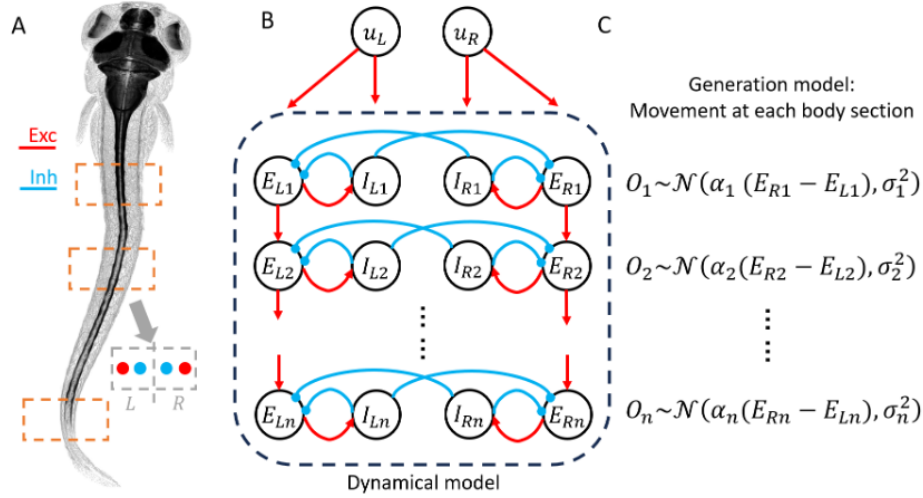


Figure 3.3: Illustration of the spinal motor network. (A) The zebrafish structure, highlighting the spinal cord and segmental neural circuits. Excitatory (Exc) and inhibitory (Inh) neurons are located bilaterally in the spinal cord. (B) Neural circuit model: Motor commands u_L and u_R from the reticulospinal system modulate excitatory (E_{L_k}, E_{R_k}) and inhibitory (I_{L_k}, I_{R_k}) neurons in each spinal segment k , shaping motor patterns. (C) Generation model: The tail movement at segment k is driven by the excitatory neuron activity difference in that segment. Here, O_k represents the active torques, and we assume $a = \alpha_1 = \dots = \alpha_n$ and $\sigma^2 = \sigma_1^2 = \dots = \sigma_n^2$.

3.2.2 Specific neural network Parameters are Required to Produce Oscillatory Movement

Now that we have designed a neural network to model neural circuits in the spinal cord, it is necessary to determine parameters that lead to expected CPG activity. That is, we want to understand conditions for the synaptic weights that ensure oscillations happen. This can be calculated by finding the eigenvalues of the synaptic weight matrix and forcing each value to have a non-zero imaginary part. Focusing on the neuronal dynamics of the first layer, we simplify the system as a linear RNN by only considering positive inputs to the ReLU-activation function. Then, we can find the eigenvalues of the synaptic weight matrix W :

$$\begin{aligned}
0 &= \det(W - \lambda I) \\
0 &= \det \left(\begin{bmatrix} -\lambda & -w'_{ips} & -w'_{con} & 0 \\ w_{ips} & -\lambda & 0 & 0 \\ 0 & 0 & -\lambda & w_{ips} \\ 0 & -w'_{con} & -w'_{ips} & -\lambda \end{bmatrix} \right) \\
0 &= -\lambda \det \left(\begin{bmatrix} -\lambda & 0 & 0 \\ 0 & -\lambda & w_{ips} \\ -w'_{con} & -w'_{ips} & -\lambda \end{bmatrix} \right) - w_{ips} \det \left(\begin{bmatrix} -w'_{ips} & -w'_{con} & 0 \\ 0 & -\lambda & w_{ips} \\ -w'_{con} & -w'_{ips} & -\lambda \end{bmatrix} \right) \\
0 &= -\lambda(-\lambda^3 - \lambda w_{ips} w'_{ips}) - w_{ips}(-\lambda^2 w'_{ips} + (w'_{con})^2 w_{ips} - w_{ips} (w'_{ips})^2) \\
0 &= \lambda^4 + (2w_{ips} w'_{ips}) \lambda^2 + (w_{ips})^2 ((w'_{ips})^2 - (w'_{con})^2) \\
\lambda^2 &= \frac{1}{2} (-2w_{ips} w'_{ips} \pm \sqrt{4(w_{ips})^2 (w'_{ips})^2 - 4(w_{ips})^2 ((w'_{ips})^2 - (w'_{con})^2)}) \\
\lambda^2 &= w_{ips} (-w'_{ips} \pm w'_{con}) \\
\lambda &= \pm \sqrt{w_{ips} (-w'_{ips} \pm w'_{con})}
\end{aligned}$$

If these eigenvalues have an imaginary part, then the system will spiral around its fixed point. Hence, in order for the model to exhibit realistic swims, we must have that

$$\boxed{w'_{con} < w'_{ips}} \quad (1)$$

That is, we must have stronger ipsilateral inhibitory connections than contralateral inhibitory connections. Since our model has no ascending projections, it follows that this is the necessary condition for the first layer to have oscillations even when there are multiple layers. Additionally, if there are oscillations in the first layer, its easy to see that there must be oscillations in the next layers.

3.2.3 Biomechanics are necessary to produce realistic swim bouts

Using our neural network design and synaptic weight conditions, we can simulate artificial swimming motion based on neural activity sent from command signals. Studying the resulting behavior can then give insights into the capabilities of our neural network and key us into what is missing to replicate locomotion.

Note that in our original assumption, the neural activity produced torques on the tail, but for now we assume that it directly produces successive differences between cumulative tail angles. Being careful to ensure equation (1) holds, we chose w'_{con} to be 10% of w'_{ips} . Our final parameters, chosen to best replicate simple forward motion, were $w_{ips} = w'_{ips} = 3$, $w'_{cons} = 0.3$, $w_{desc} = w_{ue} = w_{ui} = 1.5$, $\tau_e = \tau_i = 0.02$, $a = 300$, and $r_e = 2$, $r_i = 8$. We then simulated the network using an artificial input of $u_R = 2$ at $t = 0.01s$ and simulated the resulting dynamics using Euler's method for 0.3s with $\Delta t = 0.001s$ 3.4.

This simulation replicated some expected tail kinematics. For example, the tail was able to oscillate back and forth until coming to rest, and there is a phase lag between each segment. However, this phase lag is not constant overtime. Observing the differences in angle amplitudes over time, it evident that these differences are nearly 0 at the beginning of motion and increase as motion decays. Hence, this model lacks a realistic delay in the onset of movement. Additionally, the tail takes a unrealistically long time to decay. The tail motion is also faily symmetric despite an asymmetric input, which is atypical even for forward swim bouts of zebrafish.

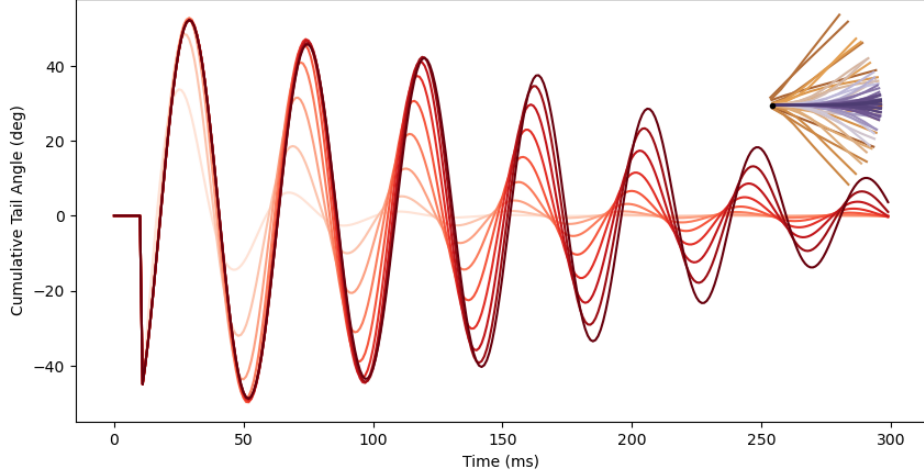


Figure 3.4: Resulting cumulative tail angles without incorporating biomechanics. Darker red color indicates angle measurements from more caudal segments. Top right inset showcases motion of tail overtime, with time varying from orange to purple.

For these reasons, we determined our neural network can sufficiently provide oscillatory motion but is in unable to replicate realistic locomotion alone. We hypothesized this is for 2 distinct reasons: (1) our model does not incorporate the interactions between the tail segments, which causes a discontinuous initiation of movement due to a lack of “internal resistance” felt between the segments, and (2) our model does not include a drag force felt by the water, which likely contributes to an unrealistically long time to decay in motion. Due to these observations, we moved towards developing a biomechanical model that takes into account both the neural activity and the bodily forces felt according to Newton’s laws of motion.

3.3 Biomechanical models

3.3.1 Model Setup

Our previous results showed the neural dynamics is capable of producing oscillatory swims but cannot replicate complex motion. Seeing the importance of the physics of the water and the tail, we determined that considering the biomechanics is important in modeling such complex behavior. We designed our model using a simplified version of past models simulating joint movement.^{10,11,16} Importantly, instead of modeling the tail positions directly, we focused on modeling the tail angles to help reduce our dynamics to a linear model.

We will now work through our biomechanical assumptions to help derive our dynamical equations using the laws of physics. First, we imagined each segment on the zebrafish as an elliptical cylinder. Each of these cylinders has a moment of inertia I_i , and the total torque on segment is given by $I_i\ddot{\phi}_i$, where $\ddot{\phi}_i$ is angular acceleration. In total, we assume there are three torques acting on the tail. The first is the *active torque* produced by the neural network, given by τ_{a_i} . Secondly, we have the *passive torque* on the tail: as a segment moves, it encounters internal resistance due to stiffness in the tail. We assume that this torque is proportional to the difference in angles between the two segments, given by the relative angle $\theta_i = \phi_i - \phi_{i-1}$. Note that, for both active and passive torques, torques acting on i^{th} segment then act in the opposite direction on the $i + 1^{th}$ segment due to Newton’s third law. Finally, we have the torque produced by the drag due to water, which we assume is linear with angular velocity $\dot{\phi}_i$.

This allows write a system of ODEs for our biomechanical model based on Newton’s laws by writing an

expression for the total torque acting on each segment:

$$\begin{aligned}
I_1 \ddot{\phi}_1 &= -\gamma \dot{\phi}_1 - k_1 \theta_1 + \tau_{a_1}, \\
I_2 \ddot{\phi}_2 &= -\gamma \dot{\phi}_2 - k_2 \theta_2 + k_1 \theta_1 + \tau_{a_2} - \tau_{a_1}, \\
I_3 \ddot{\phi}_3 &= -\gamma \dot{\phi}_3 - k_3 \theta_3 + k_2 \theta_2 + \tau_{a_3} - \tau_{a_2}, \\
&\vdots \\
I_n \ddot{\phi}_n &= -\gamma \dot{\phi}_n - k_n \theta_n + k_{n-1} \theta_{n-1} + \tau_{a_n} - \tau_{a_{n-1}}.
\end{aligned}$$

Here, γ represents the coefficient of water resistance and $k\theta_i$ represents the passive torque generated by the i^{th} segment moving. We refer to each k_i as the *stiffness constant* acting on the i^{th} segment.

Notice that due to Newton's third law, each layer has been coupled to include differences in the passive and active torques. For instance, θ_2 applies a negative passive torque on the 2^{nd} segment and a positive passive torque on the 3^{rd} . Additionally, these passive torques are given in terms of relative angle, but we would like our model to be in terms of only cumulative angles. Using the fact that $\theta_1 = \phi_1$ and that $\theta_i = \phi_i - \phi_{i-1}$ for $i > 1$, we can rewrite our equations:

$$\begin{aligned}
I_1 \ddot{\phi}_1 &= -\gamma \dot{\phi}_1 - k_1 \phi_1 + \tau_{a_1}, \\
I_2 \ddot{\phi}_2 &= -\gamma \dot{\phi}_2 - k_2 \phi_2 + (k_1 + k_2) \phi_1 + \tau_{a_2} - \tau_{a_1}, \\
I_3 \ddot{\phi}_3 &= -\gamma \dot{\phi}_3 - k_3 \phi_3 + (k_2 + k_3) \phi_2 - k_2 \phi_1 + \tau_{a_3} - \tau_{a_2}, \\
&\vdots \\
I_n \ddot{\phi}_n &= -\gamma \dot{\phi}_n - k_n \phi_n + (k_{n-1} + k_n) \phi_{n-1} - k_{n-1} \phi_{n-2} + \tau_{a_n} - \tau_{a_{n-1}}.
\end{aligned}$$

We can then write a vectorized form for the biomechanics. Let D be the $n \times n$ matrix given by

$$D = \begin{bmatrix} 1 & 0 & \dots & 0 & 0 \\ -1 & 1 & \dots & 0 & 0 \\ 0 & -1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & -1 & 1 \end{bmatrix}$$

and let K be the diagonal matrix of stiffness coefficients $k_1, k_2, \dots, k_n$. Note then that $\theta = D\Phi$. We can then represent the biomechanics by the differential equation:

$$\mathbf{I}_{\mathbf{CM}} \cdot \ddot{\Phi} + \gamma \dot{\Phi} + DKD\Phi = D\tau_{\mathbf{a}}.$$

Here, $\mathbf{I}_{\mathbf{CM}}$ is a vector of the moment of inertias $I_1, I_2, \dots, I_n$ of each tail segment. To make calculations easier, we can put this equation in terms of the variables $\beta_i = \gamma/I_i$ and $\alpha_i = k_i/I_i$ and get

$$\ddot{\Phi} + B\dot{\Phi} + DAD\Phi = D\tau_{\mathbf{a}}.$$

where B is the diagonal matrix of β 's and A is the diagonal matrix of α 's

3.3.2 Eigenanalysis of Biomechanical Models Gives Conditions to Produce Oscillations

Our biomechanical model design is based on simple equations governing the dynamics of the tail motion according to some assumptions. Like our neural network, now that we have a model design we need to find conditions for which the equations will produce oscillations in movement. These conditions therefore give parameters that are dynamically plausible, and will aid us in the inference of our model.

Similar to our neural network, we determine conditions on the parameters by eigenanalysis. However, to perform this analysis we need our equations to be in a linear form. To do this, we rewrite our equation in state-space matrix form using the state $\mathbf{x} = \begin{bmatrix} \Phi \\ \dot{\Phi} \end{bmatrix}$. Then we can write

$$\dot{\mathbf{x}} = F\mathbf{x} + M\mathbf{u}$$

where F and M are the block matrices

$$F = \begin{bmatrix} 0 & I \\ -DAD & -B \end{bmatrix}, \quad M = \begin{bmatrix} 0 \\ D \end{bmatrix}$$

This gives a linear system that allows us to study if the system exhibits oscillations like with the neural dynamics. We can find the eigenvalues of F using:⁴⁵

$$\begin{aligned} 0 &= \det(F - \lambda I) \\ 0 &= \det(-\lambda I) \det((-B - \lambda I) - (-DAD)(-\lambda I)^{-1} I) \\ 0 &= (-\lambda)^n \det(-(B + \lambda I) - (1/\lambda)(DAD)) \end{aligned}$$

Note that B is diagonal and DAD is upper triangular, hence the determinate of the remaining expression is just the product of the diagonal. This gives

$$\begin{aligned} (-\lambda)^n \prod_{i=1}^n (-\gamma/I_i + \lambda - k_i/\lambda I_i) &= 0 \\ \prod_{i=1}^n (\lambda^2 - (\gamma/I_i)\lambda - k_i/I_i) &= 0 \end{aligned}$$

Finally this gives the solutions for the eigenvalues of F :

$$\boxed{\lambda_{2i,2i+1} = -\frac{\gamma}{2I_i} \pm i\sqrt{\frac{k_i}{I_i} - \left(\frac{\gamma}{2I_i}\right)^2}}. \quad (2)$$

This holds given $4I_i k_i \geq \gamma^2$. That is, the tail stiffness has to be sufficiently large compared to the water resistance in order to exhibit oscillations.

3.3.3 Resulting Temporal Frequencies May Lead to Inaccurate Motion

We now turn to understanding the equations for dynamically plausible parameters to gain insights into how realistic motion behaves under our biomechanical model. The eigenvalues we derived tell us the system is governed by exponential decay rates

$$q_i = -\frac{\gamma}{2I_i}$$

and frequencies

$$\omega_i = \sqrt{\frac{k_i}{I_i} - \left(\frac{\gamma}{2I_i}\right)^2}.$$

Of course, each eigenvalue does not affect each tail segment individually. However, it can be shown that each eigenvector only has consecutive zero coordinates down the tail. Specifically, the $2i$ and $2i + 1^{th}$ eigenvalues only affect the tail position and velocity of the i^{th} tail segment and beyond. Mathematically, this is given by the $2i$ and $2i + 1^{th}$ eigenvectors only having non-zero coordinates in the i^{th} through n^{th} coordinates and the $n + i^{th}$ to $2n^{th}$ coordinate. This means we can order and speak of the model's eigenvalues as going "down the tail" in the sense that eigenvectors cumulate down the tail and do not affect tail segments beforehand.

These equations tell us what our biomechanical model implies about swimming motion. From the decay rate equation, we observe that the rate of decay is inversely proportional to the tail segment's mass. This may make sense, as more caudal tail segments have more surface area to provide thrust, meaning it has to push against the water strongly. Additionally, temporal frequencies w_i of tail motion seem to be determined entirely according to the tail stiffness constants k_i . This suggest that the zebrafish is constrained to move at a certain temporal frequency given their tail structure, and to move otherwise would require an oscillating torque drive to counteract the frequencies of the tail. Modeling using constant tail stiffness constants, therefore, would imply that the zebrafish must regulate their tail beat frequency, and hence resulting swim speed, from neural activity generating active torques alone.

We were also interested if we could use properties of real swimming data to infer how our biomechanical model would perform via these equations. Plots of real swim bout data suggests that much of the tail moves at the same temporal frequency. This can be replicated by the model by keeping k constant throughout the tail. However, rewriting the models equations to keep cumulative angles for a specific tail segment on the leftmost side gives, for some tail segment n :

$$I_n \ddot{\phi}_n + \gamma \dot{\phi}_n + k_n \phi_n = (k_n + k_{n-1}) \phi_{n-1} + k_{n-1} \phi_{n-2}$$

To solve this, we can view cumulative angle motion from previous segments (ie. ϕ_{n-1} and ϕ_{n-2}) as driving forces. If the driving force moves at the same temporal frequency, then the resulting equations exhibit resonance. Hence, the amplitudes of the angles will grow largely over time. This means that torques that are too large are likely to cause the tail angle amplitude to grow unwieldy before it is able to decay.

3.3.4 Broadly Distributed Motor Inputs are Likely Necessary

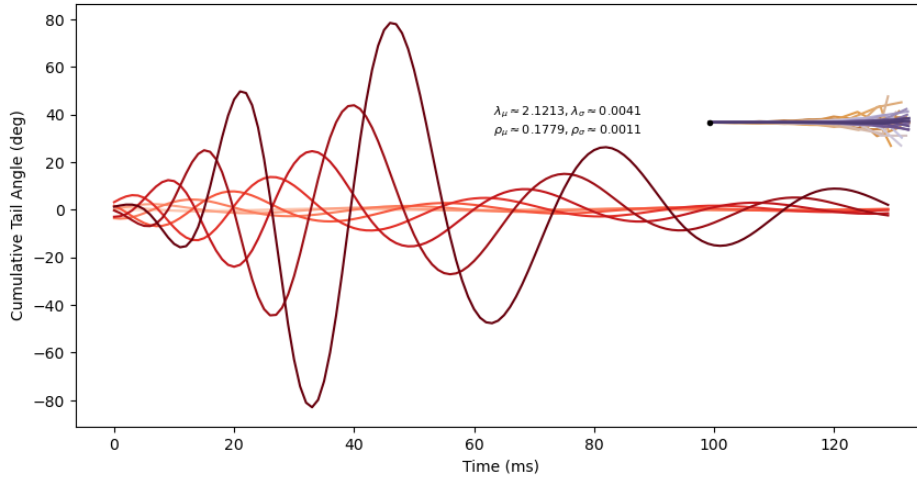


Figure 3.5: Resulting cumulative tail angles with incorporating biomechanics. Here, we have plotted from $t = 30\text{ms}$ to $t = 160\text{ms}$. Darker red color indicates angle measurements from more caudal segments. Top right inset showcases motion of tail overtime, with time varying from orange to purple. Metrics displayed indicate mean and variance of spatial wavelength λ and temporal phase lag ρ .

Now that we have designed a possible neuromechanical model and studied its parameters, we were interested in seeing the differences in this model's performance compared to a neural network alone. Similar to our neural network simulation before, we simulated tail movement with our combined neuromechanical model by sending a single sparse left motor control input to examine its effects. We used mostly the same neural network parameters as in 3.4, but notably now use the proportionality constant $a = 30$ to account for modeling torques and no longer modeling angles directly. For our biomechanics, we used the parameters $I_i = 0.01, k_i = 1000, \gamma = 5$. The results are shown in Figure 3.5.

Comparing with the real tail kinematics shown in Figure 2.2, we notice that both have characteristic oscillatory behavior, with peak angular movement at the distal tip, and last for similar amounts of time. One striking difference is the lack of movement in non-distal segments in the artificial data compared to the realistic data. This might be because the artificial data sends an input at the top of the spine, but as this signal propagates down the spine and eventually reaches the distal tip, it is comparatively weak, and hence applies a weak torque. Therefore, the tail does not move that much compared to the neural network alone. However, due to the internal coupling of the segments, moving the beginning segment just a little eventually propagates to large motion in the distal tip. Hence, even with little motion in beginning segments, the end of the tail moves substantially. This further confirms our earlier observations that a constant k_i throughout the tail would lead to resonance, and may highlight that this value is not constant along the tail. If it were, and one wanted to have bigger movement in the beginning segments, it would likely lead to uncontrollable

motion in the distal segments. Additionally, this distal movement is delayed, as the peak angular movement happens in the second half bout rather than the first, which is usually not seen in normal zebrafish swim bouts.

We believe these results indicates that input into the first layer is not enough to fully replicate realistic locomotor behavior. In order to have the distal tip reach peak angular movement earlier, it needs direct motor input from the head. Additionally, in order to move proximal segments more without causing the distal tip to move out of control, there likely needs to be motor inputs all along the tail to help proximal segments move more and keep distal segments stable. Hence, the current RNN and biomechanics are not enough to efficiently generate most zebrafish tail kinematics. Indeed, some zebrafish swim bouts such as J-turns are much more complex and asymmetrical than the one artificially generated. In the future, we believe it will become necessary to include motor control inputs that are more broadly distributed across the tail in order to replicate these behaviors. This would increase the complexity of our current RNN model, but would allow for more control in creating realistic behavior.

The comparatively large phase lag of our simulation additionally confirms the lack of segmental control. While both the wavelength and the phase lag have relatively low variances, indicated a constant spatial wavelength and intersegmental temporal phase lag similar to that of natural swimming behavior, the phase lag is likely too high to be realistic. This indicates that the biomechanical parameters may need to be further tuned to allow for faster movement, or that the neural network needs to be further improved upon to better regulate a faster phase lag, such as including more connections between segments.

3.3.5 Biomechanical Model Parameters Have Large Variable Effects on Tail Motion

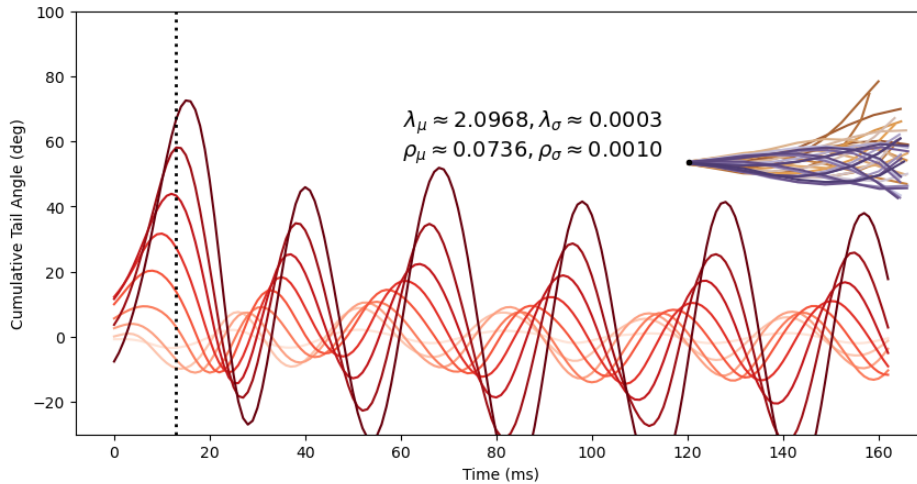


Figure 3.6: Example rightward forward swim bout used for comparison against simulations. Darker red color indicates angle measurements from more caudal segments. Top right inset showcases motion of tail overtime, with time varying from orange to purple. Dashed line gives marked time point where initial positions and velocities were calculated from. Metrics displayed indicate mean and variance of spatial wavelength λ and temporal phase lag ρ .

While our neuromechanical model may not be complete, it is still valuable to see what insights we can infer about real swimming motion using our simplified model. Thus, we moved towards attempting to match our biomechanical model as close to realistic movement as possible. However, our model has many undetermined parameters. Before we attempt to fit our model to locomotor data, it would be helpful to understand the large parameter space and how different parameters affect resulting motion. Therefore, we first conducted exploratory data analysis on simulations of various parameter sets under various conditions to understand which parameter combinations captures realistic swimming motion.

To explore the large parameter space for stable swim dynamics, we constrained the model based on the eigenvalues and expressed parameters in terms of the underlying decay rate and frequencies rather than the

parameters themselves. First, we set equal moment of inertia along the tail using $I_i = 1$. Then, to test if tail stiffness should change along the tail, we ran simulations with different varying geometric progressions of frequencies along the tail. For example, a frequency ratio of 0.5 means that the k parameter is changed so that frequencies given by the eigenvalues decrease by a factor of 0.5 down the tail. Our base frequency in all simulations was set to $\omega = 0.1$, and we used $\gamma = 50$.

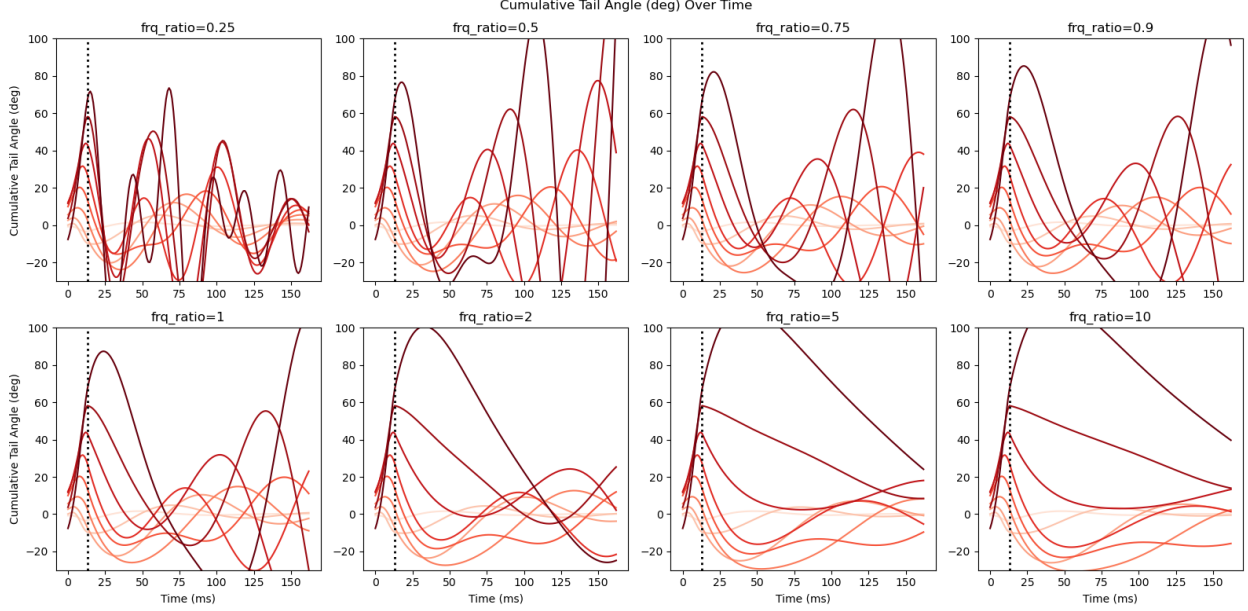


Figure 3.7: Different biomechanical simulations each run with the same initial position and velocity and letting come to rest. Darker red color indicates angle measurements from more caudal segments. Dashed line gives marked time point where initial positions and velocities were calculated from and simulation begins.

We tested how different parameter sets reacted given a realistic initial tail position and velocity. These initial values were taken from a real rightward forward swim bout (Figure 3.6). After $t = 13\text{ms}$, each simulation was then let to come to rest. Under each parameter set, the results produced oscillatory motion (Figure 3.7). However, it is evident that frequency ratios above 1 produce dynamics that are too slow as the tail comes to rest. In contrast, frequency ratios below 1 produce dynamics that seem highly variable as the tail comes to rest. This variability increases down the tail, as an initial small movement from an early tail segment propagates down to a large movement at the end of tail. This behavior seems unrealistic, as no torques are being applied to the tail in these simulations. Thus, such tail movements seem to whip around widely despite no new energy being added to the system.

This effect however is mitigated when non-constant torques are used. Having moment of inertia decrease by a factor of 0.5 down the tail gives tail dynamics that, although still somewhat variable over time, relaxes more easily (Figure 3.8), giving evidence that the moment of inertia is non-constant and decreasing.

From these new simulations, we also looked at measures of wavelength and phase lag. All simulations had relatively similar spatial wavelength to the example data, with relatively low variance. However, each simulation’s wavelength variance was higher than that of the example. The phase lag is also much larger in most simulations than the example data (except for frequency ratio 0.5), all with higher intersegmental variance. This indicates that it may be harder to get matching phase lag by tuning biomechanical parameters alone, or at least through tuning the frequency ratio. This may give evidence that the phase lag is regulated through neural activity even when the tail is coming to rest, or that our biomechanical model needs further improvement.

Finally, we ran different simulations in which the tail is at rest and a uniform torque is applied for the first 2 milliseconds. In these simulations the tails seem to either “curl up” widely or come to rest too slowly, depending on the frequency ratio. However, a frequency ratio of 1 has a relatively realistic phase lag ($\rho_\mu = 0.0725$) and low intersegmental variance ($\rho_\sigma = 0.0019$), possibly indicating that frequency ratios

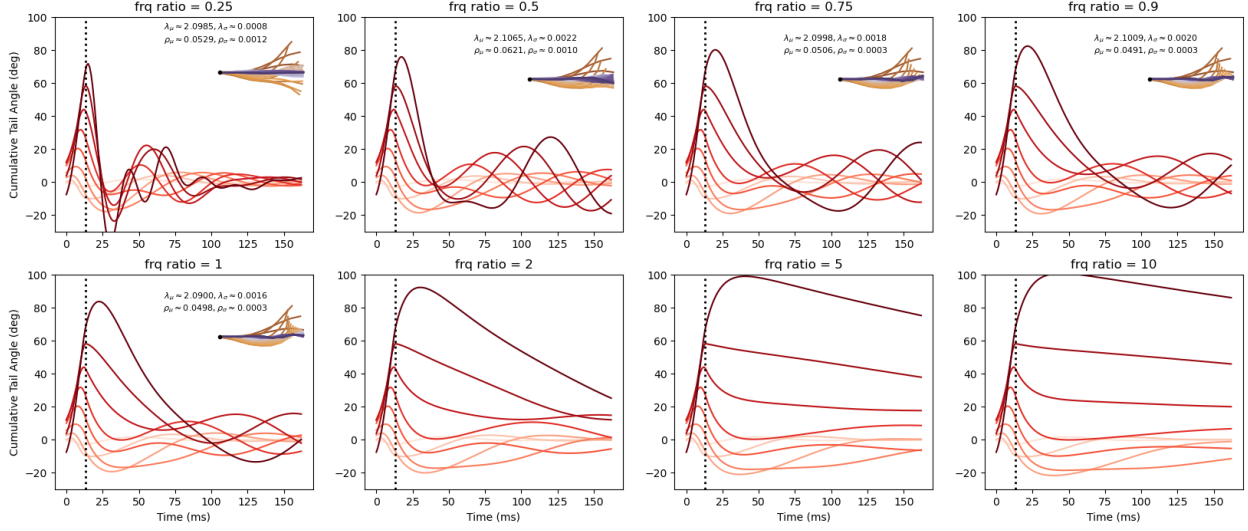


Figure 3.8: Different biomechanical simulations each run with the same initial position and velocity and letting come to rest, this time with moment of inertias decreasing by 0.5 down the tail. Darker red color indicates angle measurements from more caudal segments. Dashed line gives marked time point where initial positions and velocities were calculated from and simulation begins. Inserts and metrics shown only for some parameter sets. Top right inset showcases motion of tail overtime, with time varying from orange to purple. Metrics displayed indicate mean and variance of spatial wavelength λ and temporal phase lag ρ .

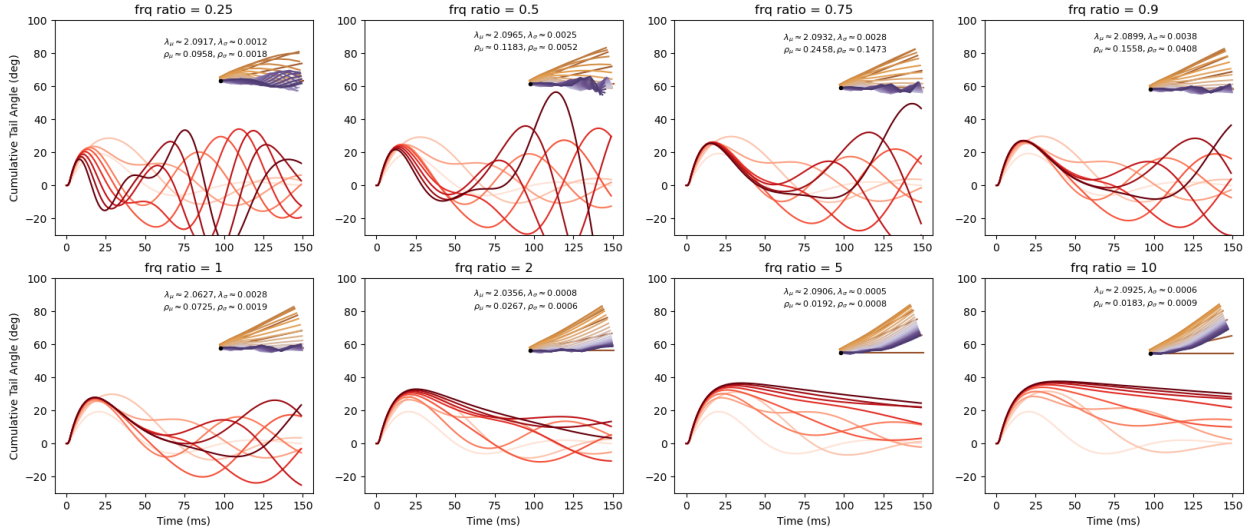


Figure 3.9: Different biomechanical simulations each run with the initial position at rest and a constant torque being applied down the tail for the first 2 milliseconds, with moment of inertia decreasing by 0.5 down the tail. Darker red color indicates angle measurements from more caudal segments. Top right inset showcases motion of tail overtime, with time varying from orange to purple. Metrics displayed indicate mean and variance of spatial wavelength λ and temporal phase lag ρ .

near 1 are more plausible. If this behavior was indeed realistic, it would indicate the biomechanics alone brings the tail quickly to rest. In addition, we see similar results for a frequency ratio of 0.25 ($\rho_\mu = 0.0958$, $\rho_\sigma = 0.0018$). This behavior by contrast indicates that further neural activity would be required to bring the tail quicker to rest after a neural input. The fact that two contrasting behaviors exhibit similarly valid measurements highlights that these metrics may not be sufficient to determine realistic biomechanics alone.

Nevertheless, it gives some indication of realistic parameters, as simulations in between these have much larger phase lags with more variance (and hence more “curling up” motion), while simulations with larger frequency ratios have phase lags that are too low (as they do not even come fully to rest).

Therefore, while it seems difficult to pinpoint realistic parameters from this analysis alone, it does give insight on how these parameters affect overall motion. Firstly, it seems clear that decreasing moment of inertias is necessary to have stable dynamics. Second, increasing temporal frequency down the tail is likely unrealistic, as the resulting dynamics decays too slowly and may not even fully come to rest. Third, no one frequency ratio seems to have stable phase lag in all simulations, indicating this may be regulated by neural mechanics as well or that our biomechanical model is not sufficient. Finally, different parameters sets can both give good metrics in terms of wavelength and phase lag, despite giving different implication on zebrafish locomotion.

3.4 Inference of biomechanics from data

3.4.1 Linear Regression Gives Intuition Behind Biomechanics Parameters

It is clear that the stiffness coefficients determine a large amount of variation in the resulting simulated locomotion under our biomechanical model. Therefore, having plausible parameters is important to understanding our model’s capabilities. We set out to infer plausible parameters by estimating from the data. To do this, we used linear regression to analyze possible parameters that could fit the data. This was done by estimating the angular accelerations using the angles and estimates of the angular velocity, and determining the γ and stiffness coefficients based on our linear biomechanical equations (Methods 7.3)

Our first question was whether $k_1 = k_2 = \dots = k_n$. Afterall, such a constraint seems to be evident given our discussion of temporal frequencies and according to our simulations. However, constraining the equation to only one k value in general performed worse then keeping them independent ($R^2 \approx 0.013286$ vs. $R^2 \approx 0.288252$), and this pattern still held when we accounted for artificial inflation due to an increase in parameters $R_{adj}^2 \approx 0.013286$ vs. $R_{adj}^2 \approx 0.288250$) likely because of our sufficiently large sample size. Additionally, the parameters in the unrestricted case were slightly more interpretable, as forcing a constant k parameter produced a negative γ value while having independence did not. For this reason, we felt that a non-constant stiffness parameter was still within possibility.

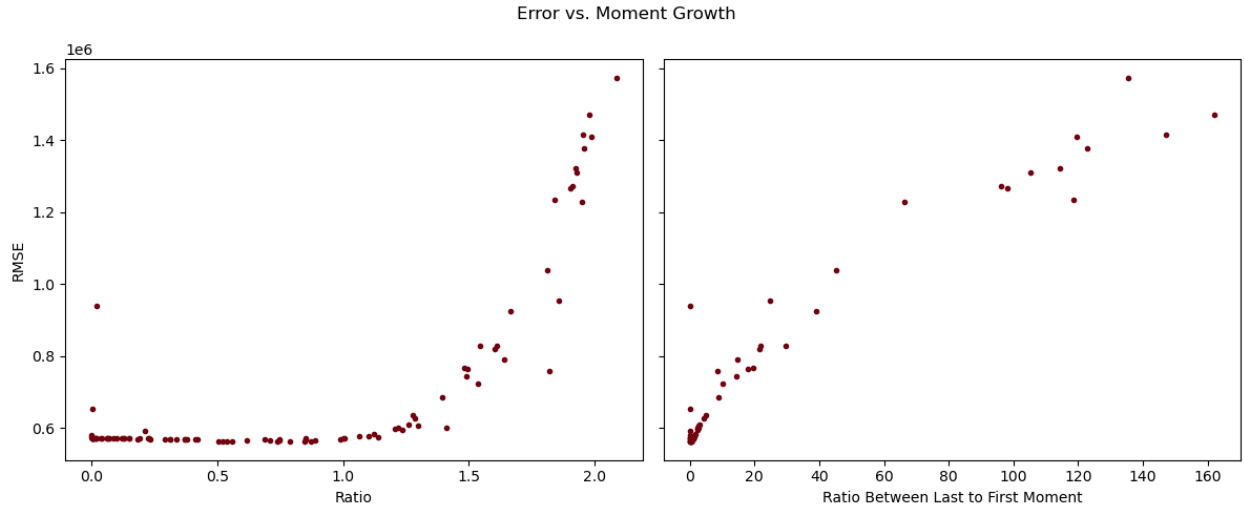


Figure 3.10: Results of CV search with 5 folds for a 100 different moment of inertia sequences. The plot on the left shows the RMSE error for each ratio r_i tested, while the plot of the right shows RMSE error in terms of the ratio between the last and first term in the moment sequence.

Next, we wanted to confirm that moment of inertia decreased along the tail. We did this by evaluating our model over various moment of inertia sequences generated based on the assumptions of our biomechanical

model (Methods 7.3). These sequences either increased or decreased according to some ratio $r^{(k)}$. Over different CV search iterations, the best moment of inertia was usually decreasing (Figure 3.10), indicating that a decreasing moment of inertia was likely better at producing optimal torques.

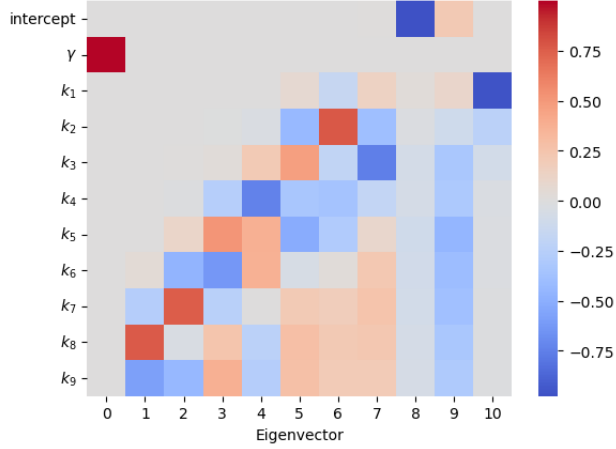


Figure 3.11: Eigenvectors of the loss function, in order of decreasing eigenvalue and hence decreasing change in the loss. Note that in this regression result, $\gamma > 0$, while most values of k_i were negative. Thus, to point towards plausible parameters, we wish to point towards increasing change of k_i with minimal change to γ . Moving in the negative directions of eigenvectors 9 and 10 hence moves towards plausible parameters.

Finally, we wanted to see if linear regression could get dynamically plausible parameters, that is, coefficients that adhere to the constraints $\gamma \geq 0$ and $\frac{k_i}{I_i} - \left(\frac{\gamma}{2I_i}\right)^2 \geq 0$. Observing the eigenvectors of the loss function showed that eigenvectors with low eigenvalues pointed in the direction of plausible parameters (Figure 3.11). This suggested we could still have dynamically plausible parameters with minimal changes in the loss, further giving evidence linear regression results could inform us about the underlying parameter space. To find such parameters, we performed constrained linear regression using quadratic linear programming optimization to add the above constraints (Methods 7.3). To ensure our estimates were not dependent on the water drag force, we ran different optimizations under the constraint $\gamma \geq \gamma_C$ for increasing levels of γ_C .

The results indicate that larger values of γ increases the measure of correlation, and this value plateaus at around 0.273 before sharply decreasing when $\gamma_C > 65$ (Figure 3.12). Looking at the value of γ as a function of γ_C , its value seems to lie around 65, independent of the value of γ_C , and the correlation decreases as γ is forced higher than this value. This may demonstrate that there are a stable set of parameters that fits the data the best under our realistic parameter constraints. Indeed, plotting the change in the stiffness coefficients overtime reveals the parameters stay relatively constant for $\gamma_C \leq 65$ (3.13). Of course, this does not mean that these are the true parameters, or necessarily result in nicely behaving dynamics. In fact, the estimated stiffness seems to increase sharply near the caudal part of the tail, which seems to contradict the results of our EDA on simulations. However, it does indicate that under the assumption there must be a fixed set of biomechanical parameters among all these types of swim bouts, there is a stable set of parameters that leads to optimal torques.

3.4.2 Regression Parameters and Torques Produce Implausible Tail Locomotion

How well behaved are the parameters we calculated through constrained linear regression? We wanted to see to what extent our inferred parameters were dynamically plausible. To do this, we used parameters from our optimization results with $\gamma_C = 65$, as this produced the greatest correlation. These parameters and the resulting torques estimated from linear regression were then used back into the biomechanical model to simulate swim bout motion (Figure 3.14).

It turns out that this evaluation resulted in fairly similar swim dynamics to realistic data, and even possessed a well-tuned phase lag. This indicates that these parameters and torques may be close to realistic

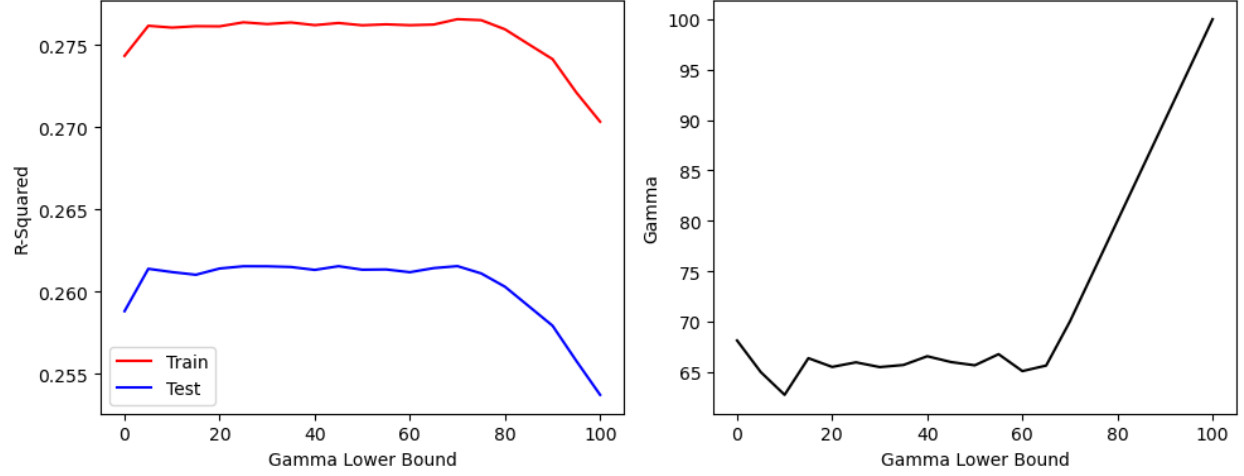


Figure 3.12: On the left, results of R^2 value for different gamma lower bounds γ_C on both training and test sets, using a decreasing moment of inertia sequence. On the right, resulting value of γ found after optimization.

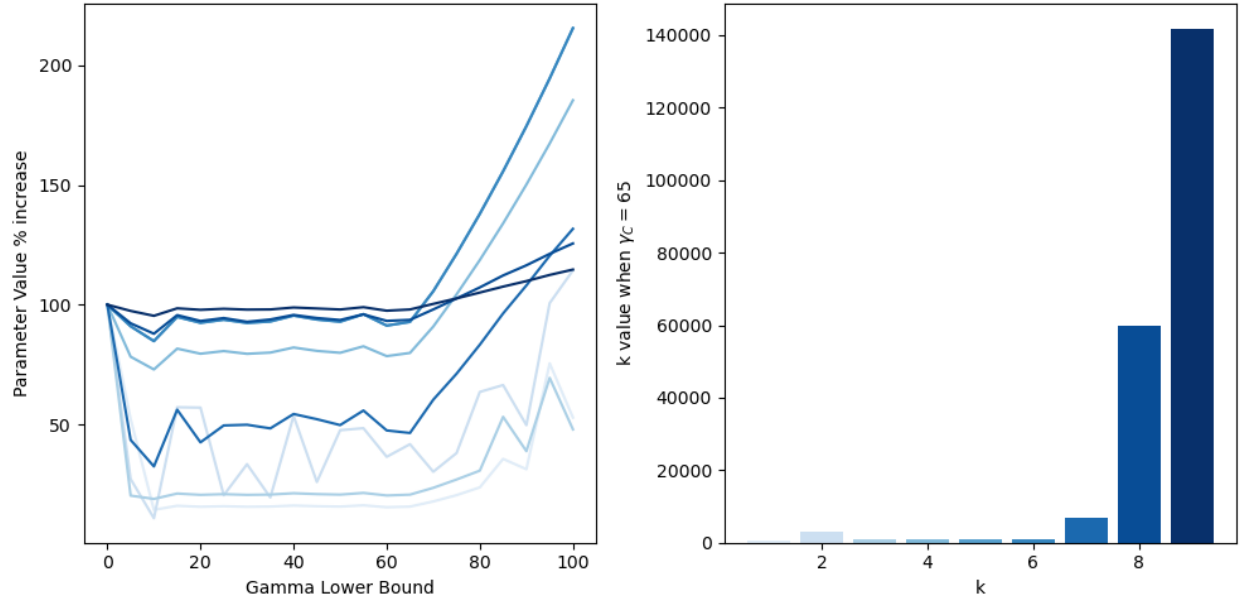


Figure 3.13: On left, percent changes in stiffness coefficients for different gamma lower bounds γ_C . On right, resulting stiffness coefficients from optimization for $\gamma_C = 65$.

values. However, these torques were not large enough to match the amplitude of the real data, and the simulated data was also far more asymmetric. Additionally, the most caudal tail segment has been omitted because it failed to stabilize overtime and exponentially increased in tail angle. From this, we determined that although regression can help inform us of the underlying parameters' properties, it is unlikely to produce parameters that provide exact stable tail dynamics.

Nevertheless, our inferred parameters and torques still does a decent job at replicating true behavior. It is therefore still meaningful to understand how our biomechanical model is able to get close to true swimming behavior by studying the estimated parameters and torques. Our estimated torque values suggest torques are largest near the end of the tail, meaning neural activation would be largest at the most caudal segments.

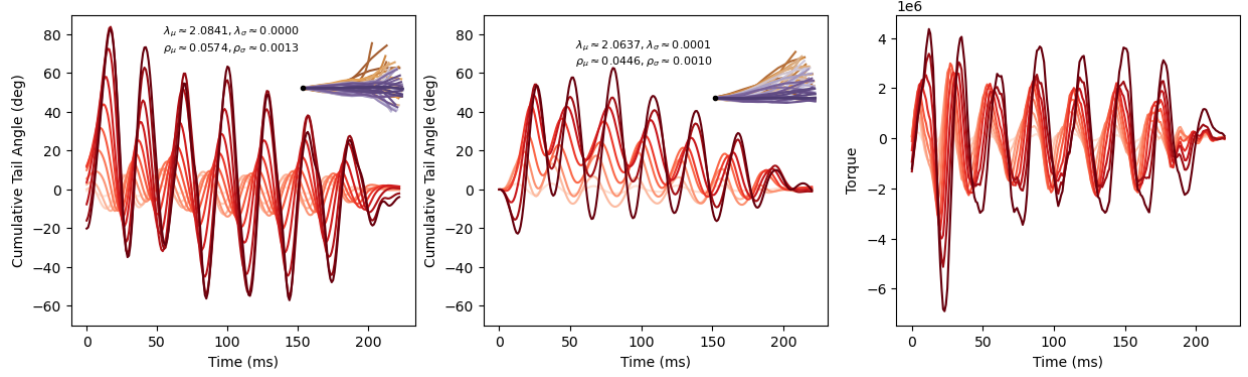


Figure 3.14: On left, example rightward forward swim bout used, repeated for convenience. In middle, resulting cumulative tail angles when regression parameters and torques are used in simulation. The most caudal tail segment has been omitted. Darker red color indicates angle measurements from more caudal segments. Top right insets showcases motion of tail overtime, with time varying from orange to purple. Metrics displayed indicate mean and variance of spatial wavelength λ and temporal phase lag ρ . On right, plot of torques from regression on swim bout which are directly given by the residuals.

This may make sense, as the fish may have to have more control near the end of the tail in order to have stable swims. Indeed, the more caudal segments seem to be the most variable in Figure 3.14, and the most caudal tail segment could not stabilize in our simulation, which suggests they may be the hardest to “keep under control” in our biomechanical simulations. Hence, more active torque is likely necessary to keep it stable. This also lines up with our previous results in which caudal tail segments usually displayed greater angle growth and amplitude. Therefore, our biomechanical model may indicate that the spinal cord would have to have greatest activity near the caudal ends of the tail.

3.5 Towards inference of neuro-biomechanical systems

3.5.1 iLQR Demonstrates Observation-Dependent Dynamics May Not Be Unique

While we have explored much of the dynamics of our neural network and biomechanical models, we have not yet thoughtfully studied how well these models work together. Additionally, much of our exploration has been done using artificial command inputs. In order to move towards inference of our combined neuromechanical model, we set out to understand what the optimal motor command inputs would look like. To study this, we inferred optimal inputs to replicate matching dynamics using iLQR,⁴⁶ a control-based inference method (Methods 7.4). This gives us a tool to understand what optimal command inputs might look like given our neural network model. To simplify, we are omitting the biomechanical model for now and modeling relative tail angles as proportional to the active torques. We are also reducing our model down to a single zebrafish layer. While not completely accurate, using this framework and focusing on a single segment allows us to see patterns that we can expect to persist in the more complicated case.

The iLQR algorithm finds optimal inputs for our dynamics based on a running cost function. This cost function includes a performance cost determined by how well the inputs generate desired observations and an energetic cost determined by the size of the inputs necessary. To determine these costs, we made some assumptions on the prior distributions of our model’s variables. We took $\Phi_t \sim \mathcal{N}(\Phi'_t, \sigma^2)$ for some σ^2 , where $\Phi'_t = a(E_R - E_L)$, and we assumed $u_t \sim \mathcal{N}(0, \sigma_u^2)$ for some σ_u^2 . These probabilistic assumptions then results in the following cost function for iLQR (see Methods 7.4):

$$\mathcal{L}(\Phi_t, \mathbf{u}_t) = \frac{1}{2\sigma^2}(\Phi_t - \Phi'_t)^T(\Phi_t - \Phi'_t) + \frac{1}{2\sigma_u^2}\mathbf{u}_t^T \mathbf{u}_t.$$

Here, we take $\sigma = 1$ and $\sigma_u = 5$.

The data fed into iLQR included a single right input early on, and then simultaneous left and right inputs later in the sequence. The first input generated simple oscillatory tail movement while the second generated

no movement due to the difference in left and right excitatory activity canceling out. We believe that since active torques only depend on the difference in excitatory activity in our current model, it can only infer the value of the difference and not the exact values. Therefore, it finds inputs that generate this difference that are optimal according to our cost function. Since our current cost function is the L^2 norm on the inputs, it results in possible negative inputs being inferred, and sometimes no inputs are inferred if the difference is zero (Figure 3.15).

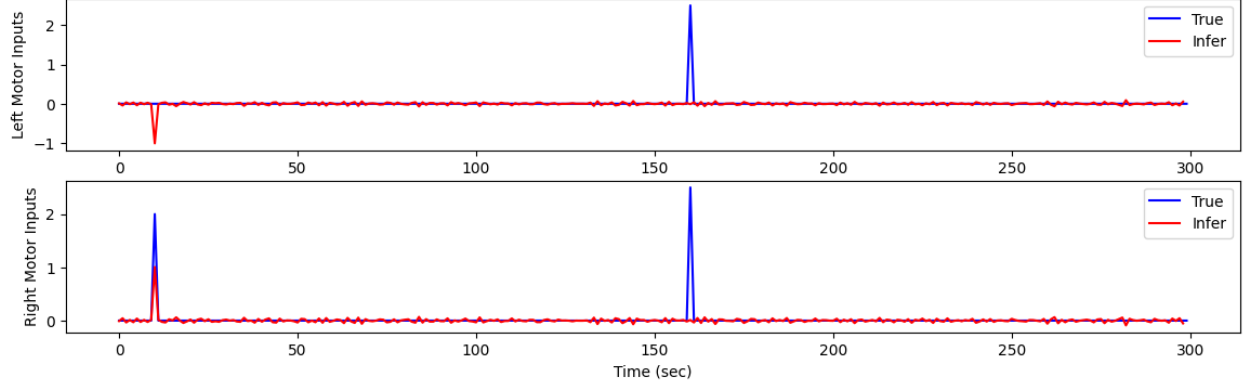


Figure 3.15: Results of iLQR inferring motor control inputs on a single layer of the neuronal RNN.

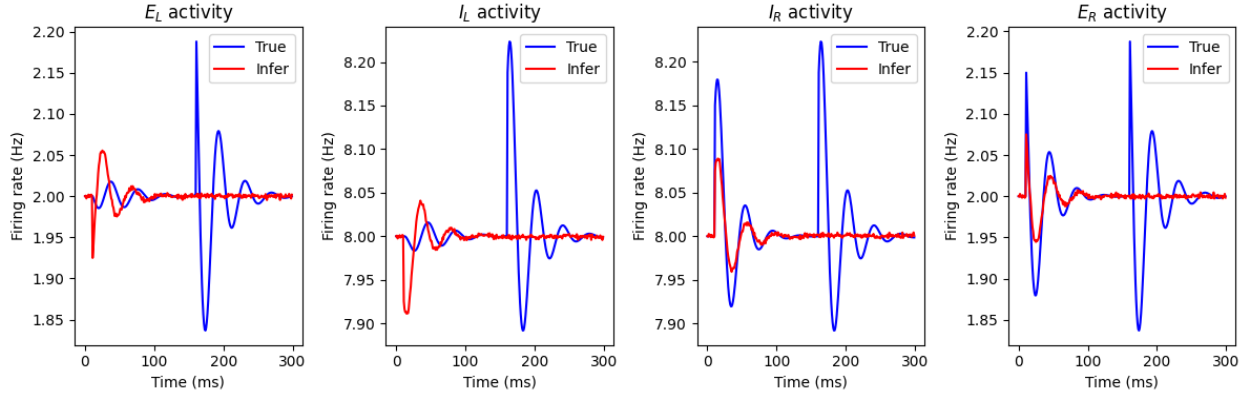


Figure 3.16: Results of iLQR inferring RNN dynamics on a single layer of the neuronal RNN.

This poses an issue as motor inputs have been shown to be excitatory rather than inhibitory, so negative inputs are less plausible. One way to remedy this is to feed information about the sum of activity into iLQR. Additionally, currently motor inputs activate both excitatory and inhibitory cells simultaneously. This could instead be changed to independent commands. The main insight is that iLQR can only infer values of $\mathbf{z}$ the model depends on, and those values may not be unique. This is will be important to keep in mind to ensure optimal values still line up with biological plausibility in our framework.

4 Discussion

4.1 Capabilities and Limitations of Our Biologically Relevant Model

Our research aims to understand how and why locomotor control is organized and optimized for various behaviors. Hence, we have sought out to model locomotion using a biologically relevant framework to understand the internal dynamics for different behaviors. Biological relevance of our models is critically important to make correct inferences and predictions on the underlying control mechanisms. Indeed, realism is important at every layer of our model, otherwise it would be impossible to interpret how certain dynamics in the brain, body, or environment interacts with the rest of the system. Previous attempts to model and understand locomotor control failed to infer meaningful aspects of internal dynamics due to not being biologically relevant.^{14,15,47} While our main goal is to understand only neuronal mechanisms, without accounting for the interplay between brain, body, and environment we may misinterpret or fail to predict correct control dynamics.^{12,48,49}

Designing a complete neuromechanical model is therefore necessary. Indeed, in our research, we quickly determined that neural networks alone are unable to account for various aspects of locomotion caused by physical interactions of the body and environment. To account for this, we used a latent dynamical framework of locomotor behavior that models descending motor inputs, neuronal dynamics, and biomechanical forces. This model allows us to better understand motor command inputs in the context of physical forces from the body and environment.

As part of this latent dynamical framework, we used biological and physical realism to design a neuromechanical model and demonstrated various results regarding its capability to generate realistic swimming locomotion. We have shown that specific conditions of our neurological and biomechanical parameters are necessarily to produce oscillations. Additionally, we have shown that our neural network model is not sufficient to produce complex locomotion. While our combined neuromechanical is more accurate, it may also still need to be improved upon, such as adding additional descending synaptic connections.

We found that our biomechanical parameters have a large influence on the resulting simulated tail motion, and explored the parameter space using EDA analysis on various parameter sets and simulated conditions. This yielded that moment of inertia down the tail likely decreases, and these results were corroborated by fitting the model to data using linear regression. A conditioned linear regression optimization further found parameters and torques that obeyed our parameter conditions and lead to dynamically plausible tail motion, albeit not exactly matching realistic data. This informed us that torques likely increase down the tail, and caudal stability may be an important issue to consider. Finally, we used control-based inference to showcase that motor command inputs are not unique under our current model framework, and only depend on excitatory differences along the tail.

However, our current neuromechanical model is subject to some limitations. For instance, our EDA analysis and linear regression analysis yielded conflicting results for the stiffness coefficients down the tail. Simulation results revealed that increasing stiffness is unlikely, while by contrast linear regression returned caudal tail stiffness that was exceptionally large compared to rostral segments. One possibility is that caudal tail segments are more variable and thus harder to estimate via regression. Another possibility is that tail stiffness is not constant for all types of locomotion, and may change with time or be subject to changes via neural activity. In fact, it is worth noting that while analysis of our temporal frequency equations seems to imply tail stiffness must be constant along the tail to keep frequencies consistent, both our simulation and linear regression results showed that non-constant stiffness coefficients were within possibility, with our simulation analysis showcasing both constant and decreasing stiffness yielding possible swimming dynamics.

However, without dynamically plausible parameters it is hard to determine if this is the case or a result of poor estimation. In fact, we must keep in mind these results are under the assumption our biomechanical model can correctly replicate realistic motion. The fact that our well optimized parameters still fail to create dynamically plausible motion may also be an indication that our biomechanical model needs to be improved.

Assuming our biomechanical model was accurate, however, why might our linear regression fail to estimate exact locomotor behavior? It is likely not true that our setup satisfies the required assumptions for linear regression. Each variable indicating angles along the tail is correlated not only in time but with each other due to internal coupling. Additionally, while the torques are normal, their mean is assumed to be given by a function in time. Indeed, analysis of our residuals gives that our model fails the linear regression assumptions.

Graphs of resulting torques were often oscillatory, a behavior that makes sense physically but obviously fails the conditions of regression as the residual plot should have no pattern (Figure 3.14). This suggests that residuals are correlated. A residual plot with no pattern, however, would fail physical assumptions. It is therefore impossible to have both interpretable torques and accurate regression through this procedure.

Nevertheless, we used this approach to get some basic intuition on the parameter space of the biomechanical model. Additionally, treating torques as residuals has merit in that zebrafish likely produce the least energy given their biomechanics. A critical difference is that linear regression is optimizing the parameters, not the torques, which would be akin to optimizing the physics of the zebrafish environment. However, organisms adapt to their environment: they typically optimize their behavior to their environmental physics rather than trying to change the environment itself. However, since regression does ensure torques are minimized, it would still allow us to look for parameters that have optimal torques, informing us of possible parameter combinations that lead to good results.

Another potential limitation is that each cumulative angle can grow without bound. In reality, tail angles are constrained by the zebrafish spine. The tricky part is that the cumulative angles are not necessarily constrained from $-\pi/2$ to $\pi/2$ (for example, consider J-turns), but the intersegment angles are. Without this constraint, many simulations produce motion that would be impossible given the rigidity of the spine. Therefore, it may be necessary in the future to update the model to constrain the intersegment angles.

How might we update our neuromechanical model to account for these various limitations? For this, it is important to understand past work on the development of biologically relevant neuromechanical models to gain insight into what our model may be missing.

4.2 Likelihood of ReLU RNNs as a model for locomotor circuits

Our model of neural activity is based on a ReLU RNN framework modeling firing rates of neuron populations in the spinal cord. How well can such a model replicate locomotor behavior, and do the limitations in our current model indicate that a new framework is necessary?

Our recurrent neural network utilized two coupled E-I oscillators in each zebrafish segment. These E-I oscillators each consist of two interneurons: one excitatory, one inhibitory. We then showed this network is able to produce oscillations given that ipsilateral inhibitory connections remain stronger than contralateral connections. However, the ability to produce oscillations does not immediately determine our network is the most biologically realistic. Despite consensus that vertebrates use these so-called central pattern generators (CPGs) in the spinal cord to produce locomotion^{39,43} there are many disagreements on their exact internal structure. Most previous work has focused on designing networks that can simply reproduce behavior without yet determining if they are actually ground truth models.

Indeed, a longstanding challenge and therefore area of focus has been designing a network that can generate multiple movements. Mechanisms for synaptic plasticity (ie. changing of neural network weights) in CPGs is unknown, but many believe this is subject to neuromodulation within the spinal cord. But this still does not indicate how the zebrafish may have such an organized locomotor repertoire. Using input variations in basic E-I oscillator can explain differences in resulting oscillations and their frequency⁵⁰ but cannot fully explain differences in bout types.

Recent work on *competitive TLNs*, or ReLU RNNs with only non-excitatory connections, has shown that these networks are capable of displaying a variety of end behaviors, including oscillations, and these end behaviors can be determined through observations of graph connectivity.^{51–53} These properties can then be extended to inhibition-dominant structures (that is, mostly non-excitatory networks) like E-I oscillators.⁵⁴ Indeed, CPGs are known to be mostly inhibitory.³⁸ From this we can study how network structure may determine dynamics^{55,56} and use this to construct E-I networks that can display multiple different dynamics to generate various behaviors.⁵⁷ For instance, one can show using these results that our two interneuron network will always generate stable dynamics back to baseline, but a three interneuron network that separated ipsilateral and contralateral inhibitory projections between two different inhibitory neurons^{16,17,36,41,58} may be able to display sustained oscillations without further input. Thus, it seems our current RNN structure can likely be expanded upon to account for complex bout types without changing the underlying model.

It’s important to note, however, that networks for swimming must not only exhibit multiple distinct swimming behaviors, but control of various features throughout each bout, such as a constant phase lag between each segment that is proportional to the swimming speed.^{35,59} Indeed, comparison between real and

simulated swims shows our model struggles to produce constant phase lag between each segment. Accounting for these aspects can promote robust and energy optimized swimming.^{59,60} While our exploration determined that our RNN model likely includes input connections down the spinal cord to replicate complex bouts such as J-turns, it has also been theorized bidirectional network connections between spinal cord segments plays a role in generating phase lag.^{38,40} Indeed, other organisms such as lamprey are known to have many intersegment connections spanning longer than just their nearest neighbor, and ascending projections are even stronger than descending commands.³⁶ In fact, evidence shows that each segment may even be structurally different or exhibit different network weights.^{38,40} This findings line up with our various results that our RNN may need to be improved to account for better segment coordination. Thus, while our current RNN model is likely robust enough to produce complex bout behavior, it nevertheless may need to be seriously structurally expanded upon in order to replicate realistic, robust behavior throughout all swimming locomotion.

Simplifying our model to ReLU dynamics is also by no means a complete description of neural activity. Indeed, oversimplification can often lead to models that are biologically implausible.⁶¹ Real neuronal biophysics in locomotor control likely exhibit various complex properties. For instance, there is evidence that locomotor circuits undergo neuromodulation in order to regulate network activity to achieve realistic phase lags and motion.^{38,49} Neurons may also have different firing properties including having distinct on-off states and the ability to slow down firing rates over time during constant excitation,^{38,39} and these properties have been shown to be important in modeling realistic motor control¹³ and creating oscillations.⁴² Thus, it may be desirable to use more realistic models of neuron firing rates.

However, previous non-linear models have been focused on only accounting for one of these properties at a time without attempting to model various aspects of biophysics simultaneously. A promising framework that may be able to capture more realistic dynamics is the Morris-Lecar model⁶² which has been shown to have useful bifurcation attributes that allow it to switch between different states (including those that exhibit oscillations) based on ion channel properties and electrical inputs.^{63,64} Networks using this model have already been utilized in model decision making²⁷⁻²⁹ and bipedal or quadrupedal locomotion,⁶⁵⁻⁶⁷ but is not yet popular in modeling swimming dynamics. These non-linear models can therefore be implemented into our framework to better account for and predict resulting complex behavior due to the biophysics of locomotor neural circuits.

4.3 Neuron control of muscle stiffness and neuromusculoskeletal models

To what extent might our biomechanical model need to be expanded upon to fully replicate realistic motion? In order to keep simplicity and interpretability, our current model is a linear implementation of the bodily and environmental interactions using Newton’s laws of motion. While interpretability is valuable, it is crucial have parameters that are still biomechanically accurate, otherwise it would be impossible to understand the internal dynamics of biomechanical control.⁴⁷ Hence, it is important to look at other potential physical interactions between the neural activity, body, and environment that may better account for realistic locomotion.

Many aspects of our model may be nonlinear and more complex in reality. For instance, using constant stiffness coefficients implies that tail beat frequency modulation is out of direct control, and inference under this assumption yielded parameters that were not completely dynamically accurate despite being optimal. Previous descriptions of biomechanics models describe stiffness as likely time-varying.⁶⁸ In fact, many previous models of locomotion exhibited behavior that was largely dependent on the choice of parameters describing stiffness, and multiple locomotor behaviors were not possible unless these parameters changed.^{41,60,69}

Our model closely resembles a linear implementation of an Ekeberg-type muscle model of lamprey swimming¹⁶ which has been implemented in other contexts, such as salamander locomotion⁴¹ and fruit-fly leg control.^{10,11} Interestingly, this model includes time-varying stiffness parameters that were under control of neural activity. In fact, these parameters were a function of the sum of neural activity on either side of each segment. Including such a term would not only make biological sense, as the zebrafish likely has control over tailsegment stiffness by changing the stiffness of its muscles, but would also allow our iLQR framework to infer both motor differences and sums. This may then allow it to infer a unique set of optimal motor command inputs.

The Ekerberg model assumes that torque is a linear function of motor neuron activation. Our model,

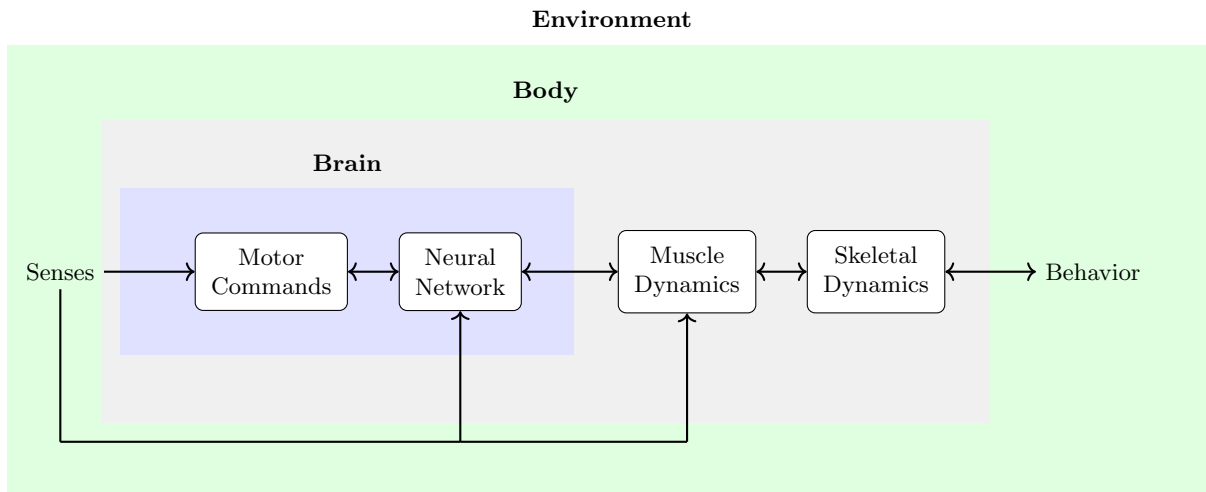


Figure 4.1: Example of a hierarchical neuromechanical framework accounting for more interactions between the brain, body, and environment, including high level control, muscle dynamics, and sensory feedback.

in comparison, assumes it is a probabilistic function, although assumes the average output is also a linear function of neuron activation. However, it has been shown that the activation of muscles by motor neurons is not a linear relationship, and depends on several factors such as velocity, muscle length, and previous activation.^{70,71} Indeed, motor neuron pools are often “recruited” in a nonlinear fashion, with stronger neurons recruited later, and this property could help explain aspects of motor inputs.⁷² To this end, many have expanded the neuromechanical framework to a neuromusculoskeletal framework, which accounts for muscle properties by modeling muscle activation via motor neurons separately from the effect the resulting activation has on the skeletal body. Models incorporating these nonlinear aspects of muscle activation lead to various locomotor capabilities, such as the ability to control oscillations.^{44,60,73} In turn, these models showcase how complex control of muscles allows for overall reduced energy costs in locomotion.⁷⁴

While these muscular properties are apparent, they are still poorly understood, and hence it is not well known how necessary it is to model these properties directly. For instance, the Ekerberg model’s linearity assumption is based on assumptions of motor neuron recruitment which, if true, would in fact allow for accurate modeling via linear implementations.⁷⁵ Whether or not these assumptions are accurate is not well determined. Therefore, including such complicated dynamics may affect inference from other aspects of biomechanics which are more well studied. Hence, for now it may be more suitable to approximate muscle dynamics as linear to more easily and accurately interpret biomechanical dynamics.

Our model also assumes the drag force caused by moving in the water is a linear function of angular velocity. The effects of fluid dynamics in the context of swimming has notoriously been a very complicated and difficult study.^{49,76} In fact, attempts to approximate fluid dynamics more accurately requires simplifications that risk being able to infer proper consequences of neuronal mechanics.¹⁷ However, modeling accurate fluid dynamics may be important as it can make the resulting locomotor behavior more optimized to the environment,⁶⁰ which may be something that zebrafish actually do. Even simple fixes to current fluid dynamics can improve accuracy, such as modeling as a function of the angular velocity squared.^{16,17,41} Yet, such implementations are still poorly understood, as much of previous modeling has focused on fluid dynamics for swimming in the context of an eel-like body, which might behave differently moving through a fluid than a zebrafish-like body. Therefore, it would be valuable to make minor adjustments in our implementation of fluid dynamics in our biomechanics and study the resulting effects on the control of swimming in zebrafish.

4.4 Towards a full brain-body-environment model

What are other brain-body-environment conditions that our model may not be accounting for? Currently, our model assumes that neural network CPGs circuits are only influenced by higher order motor input commands. This follows the classic assumption, where it has historically been thought that CPGs

are characteristically independent of sensory control due to eliciting motion even when sensory command information is removed.^{32,35,43,77}

While this may be true, such a model cannot explain how locomotor circuits *in vivo* are regulated based on context^{38,43} and most current models theorize CPGs are regulated by the environment via proprioceptive inputs.^{4,6,8,9,16,32,43,49,59,60,68,78} Such models are not only more accurate but also give insights into how animals may interact with their environment. Thus, it is evident that feedback control from the environment into CPG dynamics cannot be ignored. In fish, this usually involves *mechanosensory* information in which the muscles send feedback inputs to neurons based on their stretching and contraction. As a muscle stretches, it excites nearby neurons that cause it to contract, and vice versa. A simple way this may be modeled, adapted from the Ekerberg model¹⁶ would be

$$E_{Ri}^{(\text{prop})} = -\eta(\phi_{i+1} - \phi_i), \quad E_{Li}^{(\text{prop})} = +\eta(\phi_{i+1} - \phi_i)$$

where $E_{Ri}^{(\text{prop})}$, $E_{Li}^{(\text{prop})}$ denote activation of right and leftward proprioceptive neurons that have excitatory connections with ipsilateral excitatory neurons and contralateral inhibitory neurons. Increasing the relative angle affectively increases stretching on the left side, causing this side to excite and the right side to be inhibited. The expected effect is this feedback keep oscillations from growing out of control, and limits the growth of relative tail angles, which our results have shown may be important factors to consider.

While zebrafish do possess such mechanosensory information, it is unknown exactly how it is connected to CGP circuits.⁷⁹ Additionally, it is thought that such mechanosensory feedback in both lamprey and zebrafish are only influential in rougher environments.⁸ Nevertheless, such mechanosensory feedback could be important to include to further understand neuronal dynamics and how they are regulated.

Our model has also been ignoring the regulation of higher order motor command inputs. These inputs, sent from the reticulo formation, receive activity from higher order areas such as the MLR, basal ganglia, and cerebellum and are further regulated and separated within the reticulospinal network.^{1,3,18} These higher order brain areas process inputs from numerous cortical areas that control specific sensory information, allowing for sensory control of locomotor behavior. A more encompassing model capturing this control may be a hierarchical framework that includes high-level control of commands using sensory information from the environment.⁷ Indeed, such models highlight the importance of realistic networks on sensory control to fix errors in movement.^{10,12,17}

However, such a large frameworks often risk smaller model components no longer being biologically relevant. Past work has usually sacrificed interpretable high-level control dynamics for better model performance by using artificial neural networks^{10,14,47} or interpretable low-level locomotor control by relying on artificial oscillation generators.^{10,36,80} Another important factor to consider is that our data comes from zebrafish that were not freely swimming in their environment and had their heads locked in place. Thus, our data likely does not reflect how sensory information such as visual or vestibular senses regulate command signals. We have hence chosen to treat these effects are controlled, allowing us to focus directly on command signals and internal dynamics and how they interact with each other.

4.5 Parameter Tuning

The biggest complication we ran into for our models was finding dynamically plausible parameters that could replicate realistic motion using our models. Inferring biologically parameters is important for the same reasons we wish to infer biologically accurate dynamics. Inability to find a well fitting set of parameters may then be indicative that our models are not well suited to completely explain locomotor behavior. However, even if a well fitted set of parameters is found, this still may not indicate that our model and model parameters are biologically relevant. The reticulospinal system itself is known to incorporate a lot of redundancy between the brain, body, and environment.^{2,18,81} Such redundancy makes it hard to validate parameters or understand the validity in our models.⁴⁸ For instance, distinct sets of parameters may be able to replicate the same behavior in a neural network, or different neural networks can both replicate the same behavior with similar accuracy. Neural network parameters would be dependent on biomechanical parameters, and hence there may be multiple combinations of both that lead to the same behavior. It is clear, then, that it may be more valuable to understand the commonalities in these models that can successively reproduce behavior. This heuristic approach would give us insights on the underlying necessities of locomotor behavior, or what aspects can be ignored.

How one arrives at their chosen set of parameters also has implications on how the model is interpreted. Previous modeling efforts typically have three main approaches to parameter tuning:

1. determining through experimental data,⁹
2. tuning manually based on visually matching desired behavior,^{13,78}
3. calculating via an optimization or learning schema,^{10,14}

or some combination of those three.^{11,16,41,47} Of course, using experimental data would be most desirable in order to fully capture realistic features, but most of the time the point of modeling is that this data is unavailable or impossible to measure. One may look towards inferring using machine learning, but notably, parameters that come from the environment are fixed, hence it is more plausible to optimize around these parameters rather than optimize the parameters themselves. Our linear regression approach was limited in interpretability of model parameters due to this caveat. However, it is important to note that in some animals and environments, environment parameters *do* change, and this leads to a predictable change in behavior that models how actual locomotion changes in different environments, such as moving between land and water.¹³ Yet for most behavior, this is not the case.

A simple solution would be manually searching and picking parameters. This method can give valuable insights into the capabilities of our model and allow us to make inferences on likely properties of the underlying dynamics. For instance, picking a set of parameters that allowed for stable, oscillatory motion in our neural network allowed us to infer that biomechanics likely plays an important role. Additionally, searching the parameter space of our biomechanical model showcased that stiffness coefficients may not be constant in time and that more complex neural control is likely necessary to regulate aspects such as phase lag. Such a method therefore allows one to get insights into what is needed to replicate behavior, or how a model is able to do so, without needing to estimate correct parameters exactly. An interesting extension of this method is one that seeks to minimize model variability. That is, calculating what set of parameters minimize changes in behavior when the parameters are perturbed.^{13,73} This method assumes biologically accurate parameters are tuned to maximize structural stability, or at least allow us to look at what the dynamics would like if structural stability was maximized.

The most robust approach is using an optimization or machine learning method in an attempt to further understand the true model parameters. Sometimes, this may even be the only option, since depending on how large the neural network model may be (for example, if one used different parameters for each segment), it may be impossible to tune manually. Hence, previous methods have used a variety of methods to find these parameters.

The first category of methods, supervised learning, involves learning parameters using knowledge of errors. Importantly, our implementation of iLQR follows this regime, and is part of an even larger framework of control theory known as optimal feedback control (OFC).⁷ The benefit of iLQR is that it incorporates multiple objectives during task-oriented learning. For example, imitation learning or backpropagation may be limited to the overall objective of simply replicating desired movement, while iLQR can simultaneously optimize for other objectives such as inputs or energy cost. Thus, iLQR offers a framework to calculate optimized inputs while also giving insights into the objectives one may need to account for during locomotor behavior. Extensions of iLQR include variational inference methods to create iLQR-VAE, which uses bayesian methods to optimize for model parameters as well^{15,37} (see also Methods 7.4). This gives us an exciting tool that we can use in the future to further gain insights into our neuromechanical models through statistical inference methods.

The second category is unsupervised learning, which does not optimize according to a predefined objective and instead tunes parameters based on learning behavior or rewards. Two common approaches involve reinforcement learning (RL) or evolutionary algorithms. These methods hence give insight into a desirable set of parameters without assuming beforehand what objective must be optimized for. Understanding the parameters afterward, therefore, can give insights into what the objective might actually be. Of course, these methods are not without their caveats, and tend to have difficulty inferring plausible parameters for very large networks due to too much variability, which is why some have opted to restrict tuning based on actual neural connections.¹²

It is also important to study how animals actually learn. Learning is known to be determined both by error-based optimization and reward-based feedback through structures such as the basal ganglia or

cerebellum.^{1,3,82} However, some of the common machine learning methods based on these strategies are likely not biologically feasible, such as exact positional matching to behavior through imitation learning, or being able to keep track of large amounts of errors during backpropagation.⁸³ Parameters can still be learned via these methods for data science purposes, but understanding a realistic learning scheme would be valuable in learning how organisms actually determine neuromechanical parameters biologically.

5 Conclusion

These comparisons highlight the general differences in biomechanical modeling, showcasing the balance between interpretability of results, reproducing actual dynamics, and making meaningful inferences about the underlying control. Our optimal goal moving forward should therefore be to keep all of these aspects in mind. Having optimal command inputs that generate tail kinematics reveals how the reticulospinal system might be optimizing these commands, and, depending on our methods and model, give insight into actual dynamics or neuronal objectives. The resulting description of motor commands will give insight in how zebrafish send inputs into their tail, and importantly, what exactly they are optimizing to do so. This would give more understanding into how locomotor systems encode complex motion into input commands and integrate latent dynamics to generate different motor behaviors in vertebrates.

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7 Methods

7.1 Zebrafish Swim Bout Data

Our model contained 2238 swim bouts from a particular zebrafish. These swim bouts were categorized as either $F, Ff, Fm, T, T1, T2, Tx$, or J . These swim bout types were determined through clustering of kinematic parameters.²¹ For instance, these groups are in increasing order of angular velocity. Importantly, each category represents a swim bout with different kinematic properties, in a sequence that roughly increases in complexity. The first three swim bouts denote forward swims, the next three denote turning swims, and the last swim denotes the characteristic J-turn. Each swim bout has a rightward and leftward lateralization, giving 14 swim bout clusters in total. Any swim bout outside of these categories (uncategorized) were discarded.

Some swim bouts recorded were exceptionally long, and had properties that changed overtime. In order to prevent this from skewing our data, we filtered all swim bouts to be of typical duration, lasting less than 500ms. The resulting 1234 swim bouts were used in our analysis.

7.2 Spatial Wavelength and Temporal Phase Lag

For each swim bout, we calculated the spatial wavelength by finding the inverse of the dominant frequency in the Fourier transform of the spatial wave at each point in time. These measurements were then averaged to get an estimate of mean wavelength λ_μ , and the variance was taken to get an estimate of the wavelength variation over time λ_σ . Phase lag was calculated by using cross-correlation between each segments y -values over time to calculate a time shift, which was then multiplied by the dominant temporal frequency calculated via a Fourier transform. Similarly, we used these values to calculate the mean phase lag across all segments ρ_μ and the intersegmental phase lag variance ρ_σ .

Outliers for both distributions were removed, as most of these swim bouts had atypical properties, such as exceptionally fast swimming speeds or multiple swim behaviors per bout.

7.3 Linear Regression

Linear Regression Setup The biomechanical model was rewritten as a function in terms of its parameters:

$$I_i \ddot{\Phi} = \gamma(-\dot{\Phi}) + k_1(-DE^1 D\Phi) + k_2(-DE^2 D\Phi) + \dots + k_n(-DE^n D\Phi) + D\tau_a$$

Here, E^i is the matrix of all zeros, except $(E^i)_{ii} = 1$. We then let $\ddot{\Phi} = \mathbf{y}$, $(-\dot{\Phi}) = \mathbf{x}_0$, and $(-DE^i D\Phi) = \mathbf{x}_i$. The first and second derivatives are then estimated by the data using finite differences. If we assume each $\mathbf{x}_i$ is uncorrelated and $D\tau_a \sim N(0, \sigma^2)$ for some σ^2 , then we can rewrite this equation a linear regression problem:

$$\mathbf{I} \cdot \mathbf{y} = \gamma \mathbf{x}_0 + k_1 \mathbf{x}_1 + k_2 \mathbf{x}_2 + \dots + k_n \mathbf{x}_n$$

This allows us to then estimate the parameters γ, k_i using linear regression.

Moment of Inertia CV search We also performed CV search across moment of inertia sequences. To simplify searching through different moments, we took different randomly generated increasing and decreasing moment of inertia sequences $I^{(k)}$. These sequences were determined using the fact that each segment is modeled as an elliptical cylinder, which has a moment of inertia that is a function of its length and width.¹⁶ If we assume constant length for each segment, and a width that scales with some by some factor r , then each moment of inertia sequence is of the form $I_i^{(k)} = a^{(k)} + b^{(k)}(r^{(k)})^i$. To generate multiple sequences, $a^{(k)}, b^{(k)}, r^{(k)}$ were each sampled from a uniform distribution between 0 and 2. We then proceeded to treat the moment of inertias as hyperparameters and performed Grid CV Search to find the lowest residual sum of squares.

Constrained Linear Regression This optimization problem was implemented using Sequential Least Squares Programming (SLSQ) from the scipy library. However, we did not want our results to be biased towards a lower value of γ . This is because the most optimal value of γ is not necessarily the most correct,

as the zebrafish can only optimize its own body physics and not the physics of the water. Hence, we ran our model under different bounds $\gamma \geq \gamma_C$, where γ_C spans from 0 to 100 with a step size of 5. We then observed how the resulting fits changed as a function of γ_C . To simplify our optimization, we constrained to only look at rightward forward bouts. We split our data using a roughly 80-20 randomized train-test data split. Additionally, to make our results robust against our initial guess for each run of SLSQ, for each γ_C , we ran 10 runs of SLSQ with different random initial guesses $p_i = (x_i + 1)B_i$, where p_i is one of the parameters, B_i is the lower bound constraint for that parameter, and $x_i \sim \text{Exp}(1.5)$, in an attempt to encourage guesses close to the boundary but also allow for larger values. The parameters that produced the minimum error for each γ_C were then chosen. To then compare the resulting measures, we used the R^2 value. By using a measure of correlation, we can get a sense on of how well these parameters fit the data and not rely on optimized error, which as previously stated may be subject to misinterpretation of “optimizing” the environment.

7.4 iLQR: Iterative Linear Quadratic Regulator

A Linear Quadratic Regulator (LQR) is used to find optimal controls for a linear dynamical system based on a quadratic cost function. Optimal inputs are found by starting at the desired final observations and working recursively backwards. Inputs are changed in each iteration according to a gradient descent-like learning rule using a running cost function. An iterative Linear Quadratic Regulator is an extension of LQR that uses quadratic and linear approximations of the cost function and dynamics respectively to make it a local LQR problem.⁴⁶

The benefit of using iLQR is it allows us to infer inputs of complex dynamical systems given a set of parameters. This allows us to study optimal inputs of our neural dynamics model, and therefore infer what inputs into our model may need to look like. Additionally, iLQR can be extended into iLQR-VAE, which builds upon the optimal inputs and uses methods of variational inference to determine optimal parameters as well as inputs.^{15,37} Such a tool would allow us to determine optimal parameters in our combined model as well as optimal inputs, eliminating the need to set or simulate either one.

To calculate our running cost function, we assume a prior on the inputs $u_t \sim \pi_\theta$ and maximize the likelihood of out inputs given our outputted torques according to our latent dynamic framework 2.2:

$$u^* = \arg \max_u \log p_\theta(u|\tau_a) = \arg \max_u \left(\sum_{t=1}^T (\log p_\theta(\tau_{a_t}|z_t) + \log \pi_\theta(u_t)) \right).$$

This gives our running cost function to be optimized by iLQR:

$$\mathcal{L}(o_t, u_t) = -(\log p_\theta(\tau_{a_t}|z_t) + \log \pi_\theta(u_t)).$$

The first term acts as a performance cost against the true outputs and the outputs produced by z_t . The second term acts as an energetic cost on the inputs u_t . This running cost function is then used to update inputs according to the backwards recursion described above until it converges to some u^* .

Note this assumes we write u^* as a single vector, with each spatial dimension expressed in order of time separately.

Extension into iLQR-VAE The iLQR algorithm can be extended to not only infer optimal inputs but optimal parameters as well. This is done by assuming we can pick a good recognition model q_ϕ by mapping observations to their optimal inputs according to a Gaussian centered at u^*

$$q_\phi(u|\tau_a) = \mathcal{N}(u^*, \Sigma_s \otimes \Sigma_t)$$

In the iLQR-VAE algorithm, we simplify this distribution further: If u^* is multivariate normal, each marginal distribution is also normal. VAE employs the “reparameterization trick”,⁸⁴ in which we instead express the inputs as a deterministic function $u_\phi(o, \epsilon) = u^*(o) + s_\phi \odot \epsilon$, where s is a vector of standard deviations (equivalent to the square root of the diagonal of $\Sigma_s \otimes \Sigma_t$) and $\epsilon \sim \mathcal{N}(0, I)$.

Our goal now is to optimize the ELBO estimate, or the estimate of how well our recognition model matches our desired distribution.

$$\begin{aligned}
\mathbb{E}_{q_\phi} \left[\log \left(\frac{p_\theta(u, o)}{q_\phi(u|o)} \right) \right] &= \mathbb{E}_{q_\phi} [\log p_\theta(o|u) + \log \pi_\theta(u) - \log q_\phi(u|o)] \\
&= \mathbb{E}_{q_\phi} [\log p_\theta(o|z) + \log \pi_\theta(u) - \log q_\phi(u|o)] \\
&= \mathbb{E}_{q_\phi} \left[\sum_{t=1}^T (\log p_\theta(o_t|z_t) + \log \pi_\theta(u_t)) - \log q_\phi(u|o) \right]
\end{aligned}$$

Using the reparameterization trick for q_ϕ gives $\mathbb{E}_{u \sim q_\phi} [f(u)] = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} [f(u^*(o) + s_\phi \odot \epsilon)]$. But this can be approximated by taking n random samples of ϵ and averaging over. Hence, we rewrite the cost ELBO estimate accordingly:

$$\begin{aligned}
L &= -\frac{1}{n} \sum_{i=1}^n \left[\sum_{t=1}^T (\log p_\theta(o_t|z_t^{(i)}) + \log p_\theta(u_t^{(i)})) - \log q_\phi(u^{(i)}|o) \right] \\
&\text{where } u^{(i)} = u^*(o) + s_\phi \odot \epsilon^{(i)} \text{ for } n \text{ random samples } \epsilon^{(i)} \sim \mathcal{N}(0, I), \\
&\text{and } z_t^{(i)} = f_\theta(z_{t-1}^{(i)}, u_t^{(i)})
\end{aligned}$$

This gives the cost function for VAE. Gradients can be calculated with respect to θ and ϕ , and these gradients are used to update the parameters using *Adam's algorithm*.⁸⁵ Changing the parameters θ then changes $u^*(o)$, which is resolved through iLQR and the processes continues until the ELBO converges. We hope to use this tool for future study of our neuromechanical models.